

Research article

A comparative analysis of car fleet efficiency evolution in Europe and Australia insights on policy influence[☆]Dimitrios Komnos^{a,b}, Robin Smit^c, Leonidas Ntziachristos^a, Georgios Fontaras^{b,*}^a Aristotle University of Thessaloniki Greece, Department of Mechanical Engineering, Greece^b European Commission Joint Research Centre, Ispra, Italy^c Transport Energy/Emission Research (TER), Australia

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ABSTRACT

Regulators worldwide introduce measures to improve energy consumption and reduce greenhouse emissions of road vehicles. The present study focuses on the impacts of passenger car carbon dioxide (CO₂) and fuel efficiency measures, attempting to evaluate their effectiveness using vehicle and fleet modelling. To obtain a robust counterfactual result, two regions are compared as a case study: the European Union (EU), where CO₂ emissions have been regulated for over fifteen years, and Australia, a region that recently introduced such a policy element. The average difference in the certified CO₂ emissions of new cars registered in the two regions increased from 50 g/km in 2018 to 60 g/km in 2021 due to accelerated electrification in the EU. To demonstrate the importance of mandatory targets, a sensitivity analysis of the 2020 EU registrations showed that lower by one-third sales of zero and low-emission vehicles (Z-L-EVs) would have resulted in seven out of ten manufacturer pools failing to meet the 2020 target of 95 g/km. The simulation framework was validated against certified values of 2021 registrations and was subsequently used to calculate real-world fleet performance to quantify the actual emissions savings. The real-world CO₂ emissions in the registered fleets for 2021 were estimated to be 143 g/km and 204 g/km, for the EU and Australia, respectively. With these results as a reference, the share of ZEVs required to meet the targets set in both regions is calculated: 51% of the new registrations in 2030 in the EU for achieving 49.5 g/km, and 60% in Australia for recently set target of 58 g/km for 2029. The respective calculated average fleet-wide real-world tailpipe CO₂ emissions were estimated to be 66 g/km (EU) and 77 g/km (Australia). As a last step, the study discusses the application of similar tools and analysis in the design of future policies.

Nomenclature			
AUS	Australia	MC + NA	Heavy SUVs and light commercial vehicles group in Australia
BEV	Battery Electric Vehicle	MPV	Multi-purpose Vehicle
CD	Charge Depleting mode of a plug-in hybrid vehicle	MRO	Mass in Running Order: the mass of the vehicle, with its fuel tank(s) filled to at least 90% of its capacity, including the mass of the driver (75 kg), fuel and liquids, fitted with

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Nomenclature			
CS	Charge Sustaining mode of a plug-in vehicle	NEDC	the standard equipment in accordance with the manufacturer's specifications
CO ₂	Carbon Dioxide	NOVC-HEV	New European Driving Cycle, i.e., light-duty vehicles emission certification protocol
			Non-off vehicle charging hybrid electric vehicle

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[☆] The views expressed in the paper are purely those of the authors and should not be interpreted as an official position of the European Commission.

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Nomenclature			
EC	European Commission	NTC	Australian National Transport Commission
EEA	European Environment Agency	NVES	New Vehicle Efficiency Standard
EU	European Union	EU _{RW,2021} dataset	On-Board Fuel and/or energy Consumption Monitoring dataset 2021
FCAI	Federal Chamber of Automotive Industries	OVC-HEV	Off vehicle charging hybrid electric vehicle (PHEV)
F0, F1, F2	Road Load coefficients that explain the vehicle resistances	PHEV	Plug-in Hybrid Electric Vehicle
GHG	Greenhouse Gas emissions	R _{CDC}	Charge Depleting Cycle Range, parameter to calculate during the WLTP certification of plug-in hybrids
GVG	Australian Green Vehicle Guide	SUV	Sport Utility vehicle
HEV	Hybrid Electric Vehicle	TER	Transport Energy/Emission Research
HVAC	Heating, Ventilation, and Air Conditioning	UF	Utility Factor, i.e., kilometers travelled in electric mode to the total kilometers travelled
ICE(V)	Internal Combustion Engine (vehicle)	US	United States of America
ICCT	International Council of Clean Transport	VKT	Vehicle Kilometres Travelled
LDV	Light-duty vehicles, i.e., cars and vans	WLTP	Worldwide harmonised Light vehicles Test Procedure, i.e., light-duty vehicles emission certification protocol currently used in the EU
MA category	Passenger cars and light SUVs group in Australia	Z(L)EV	Zero (and Low)-emission vehicle. Vehicles with certified CO ₂ emissions below 50 g/km.

1. Introduction

The Paris Agreement set out the objective to keep the global average temperature increase well below 2 °C above pre-industrial levels. To achieve this goal, greenhouse gas emissions (GHG) reductions in all sectors are crucial¹. Road transport and, in particular, cars are responsible for a substantial share of carbon dioxide (CO₂) emissions in the European Union (EU) (Sporkmann et al., 2023) and worldwide (Transport Energy/Emission Research, 2019; Silitonga et al., 2012; Augustus De Melo et al., 2018; Deng et al., 2024; Karplus et al., 2013). Sustainable low-carbon mobility is thus a high policy priority globally². Many factors influence CO₂ emissions of the transport sector, e.g. economic activity, fuel efficiency, the number and the characteristics of

vehicles, availability of public transportation, and cost of purchasing and using a car (Minarta and Ko, 2024). Governments can influence, to a certain degree, these factors via taxation, subsidies, and control policies (Ptak et al., 2024). Tax exemptions or subsidies are a powerful tool to incentivise consumers to adopt innovative low-emission vehicle technologies such as electric vehicles (Yan, 2018; Chien et al., 2021). Fuel Efficiency, GHG, or CO₂ emission standards, which impose targets or limitations on fleet-level emissions, are more direct measures.

Currently, more than 85% of all vehicles sold worldwide are covered by some form of Fuel Efficiency or CO₂ target, applied at the fleet level (Australian Government, 2024). Regulated metrics are CO₂ and/or GHG emissions and fuel consumption (or fuel economy). Since these terms are interrelated, the terms CO₂ standard and CO₂ targets are adopted from this point on. The manufacturers' compliance with the target is assessed annually, and it is applied only to the newly registered vehicles of that year (the second-hand market is not considered). Also, the target is not the same for the whole fleet and every manufacturer, but it is based on attributes such as the vehicle footprint, mass, or mass classes³. CO₂ targets operate in tandem with vehicle certification. The CO₂ certification consists of several different tests, with the main two being (1) the vehicle resistance determination and (2) the emissions test in the laboratory, where the emissions and consumption are measured while the vehicle is driven over a predefined speed profile resembling urban, rural, and highway conditions. Australia⁴ uses the New European Driving Cycle (NEDC), and the EU⁵ and China (Zhang et al., 2023) did so until 2020. The NEDC is criticized for lacking real-world representativeness, resulting in a growing CO₂ gap between the certification and driving under real-world conditions (Fontaras et al., 2017; Tietge et al., 2017). To address this issue and reduce the discrepancies between the different certification cycles, a global, harmonised certification protocol for determining the levels of pollutants, CO₂ emissions, fuel consumption, and electrical range of light-duty vehicles (LDVs) was developed, the Worldwide Harmonised Light Vehicles Test Procedure (WLTP). The EU adopted WLTP in 2017, and other countries will follow.

The EU is a longstanding regulated example for passenger cars. In 2009, the EU decided on an NEDC-based fleet-wide mandatory target of 130 g/km starting from 2015 to 95 g/km from 2020 onwards. Starting from the new fleets registered in 2021, the mandatory CO₂ targets are assessed based on WLTP. Regulation (EU) 2019/631 introduced reductions on the fleet-wide targets to apply for the period 2025–2029 and 2030–2034, which were determined by applying reduction percentages to the 2021 emission targets. The reductions result in an EU fleet-wide target of 93.6 g CO₂/km and 49.5 g CO₂/km for the corresponding periods (Suarez et al., 2023a). To ensure the representativeness of the vehicle certification in real-world conditions and the actual improvements seen in the passenger cars fleet, additional measures have been added to the (EU) 2019/631. Starting in 2021, new vehicles registered in the EU are mandated to have on-board fuel and/or energy consumption monitoring (OBFCM) devices. The European Commission (EC) uses these data to monitor the evolution of the real-world divergence from the WLTP-based certification. Another measure taken since 2023, Regulations (EU) 2023/2866 and (EU) 2023/2867, is the verification of the CO₂ emissions of vehicles in service (Pavlovic et al., 2024). Vehicles from a representative sample of newly registered vehicles undergo extended testing, including the repetition of the certification protocol. This procedure is performed to verify the correctness of the declared CO₂

³ ICCT, 2017: 2017 Global update: Light-duty vehicle greenhouse gas and fuel economy standards. (<https://theicct.org/publication/2017-global-update-light-duty-vehicle-greenhouse-gas-and-fuel-economy-standards/>).

⁴ Australia will adopt Euro 6d emission Standards for newly approved vehicles in Dec 25 and all new vehicles from July 2028.

⁵ Regulation (EU) 2019/631. The EU introduced the WLTP based protocol in their certification system in 2017, but continued to base their CO₂ emission targets on the NEDC cycle until 2020.

¹ Official Journal of the European Union, L 282, October 19, 2016.

² At COP28, UNECE and partners highlight need to decarbonize inland transport and how UN tools and legal instruments can help.

and fuel consumption values, and the absence of strategies to artificially improve the vehicle's performance during the certification procedure. Together with the enhancement of the regulatory framework, in 2023, the EU amended the regulation adding a fleet-wide target equal to a 100% reduction of the target in 2021 for 2035 onwards.

Contrary to the EU, a partially regulated region is Australia. The Federal Chamber of Automotive Industries (FCAI) is the peak representative organization for companies that distribute light-road vehicles in Australia. The FCAI developed voluntary CO₂ standards⁶ with a methodology similar to the EU: the target of each manufacturer is calculated according to their sales-weighted vehicle mass. In contrast to the EU mandatory targets, where no compliance with the targets results in paying an excess emissions premium, the voluntary schema did not foresee any penalty. In 2019, Transport Energy/Emission Research (TER) analysed the trends in the average CO₂ emissions of passenger cars registered in Australia from 2005 to 2018 and compared them to those of the EU, the US, and Japan (Transport Energy/Emission Research, 2019). The study observed that Australian cars had not reduced CO₂ emission rates as fast as the other regions, attributing it to the lack of mandatory Standards and the strong and sustained growth in the sales of large and heavy vehicles. In 2024, Australia voted for the first mandatory CO₂ standards, starting to apply in the 2025 registrations. The fleet-wide target for passenger vehicles and Sport Utility vehicle (SUVs) registered in 2025 will be 141 g/km, progressively reduced to 58 g/km for 2029 (Australian Government, 2024; Parliament of Australia, 2024).

Mandatory CO₂ standards and targets are generally considered an effective strategy for rapid improvement of the fuel economy of motor vehicles (Silitonga et al., 2012; Sen et al., 2017; Pasaoglu et al., 2016; Atabani et al., 2011; Mahlia et al., 2013). Studies use a variety of methods and data to assess the evolution in the efficiency of the car fleet and the connection to the CO₂ targets or mix of CO₂ policies. Zhang et al. (2023), performing a comparative analysis of the policy packages of different regions, suggested that research and development in low-carbon technologies, the standardisation of the CO₂ certification and the targeted carbon reduction policy are important for successful CO₂ standards implementation. Zhao et al. (2019) assessed China's CO₂ regulation and BEV deployment policies adopted by the state using a bottom-up modelling framework; with 40% of the sales being BEVs in passenger cars, the whole passenger cars fleet is expected to have an average fuel consumption of 3.2 l/100 km in 2030. Fritz et al. (2019) used the official EU monitoring datasets that contain the vehicle sales and the certified CO₂ emissions of passenger cars to extrapolate the fuel economy trends to forecast the needs for the BEVs to meet the targets set by the CO₂ standards. To analyse the interactions from different policy mixes, several studies (Lam and Mercure, 2021; Pourmatin et al., 2024; Guo et al., 2022) have developed agent-based models, which are tools for modelling sociotechnical systems. In their study, Lam and Mercure (2021) concluded that there are reinforcement effects between the EV mandate and all of the other policies. Sen et al. (2017), developed an agent-based model to estimate the potential future market shares of BEVs by applying different policy scenarios for the United States (US). Pasaoglu et al. (2016) applied scenarios to analyse the technology transition in the EU passenger road transport, considering the EU CO₂ targets together with economic factors. Similar methodologies have been used to simulate scenarios to understand what mix of policies, incentives, and infrastructure would be needed for Australia's passenger cars fleet to become carbon neutral by 2050 (Broadbent et al., 2022a, 2022b, 2024). Such studies are important for designing regulations, assessing infrastructure needs, and understanding socioeconomic interactions. However, these studies either omit or make simplified

assumptions about fleet composition, characteristics, and particularities and how they affect an important factor: the real-world influence on CO₂ emissions.

The actual vehicle performance under real-world conditions can potentially dilute the CO₂ policies due to the growing gap between certified and real-world fuel and energy consumption values. This became more evident with the low-emission potential of plug-in hybrid electric vehicles (PHEVs), a technology that could serve as a bridge during the transition to full electrification. Few studies have accounted for the real-world influence on fuel economy. Tietge et al. (2017) applied regression to the certified values and a few technical vehicle characteristics and explained 87% of the variance in real-world fuel consumption from such user-submitted data. Missing such real-world indication for Australia, TER reconstructed the real-world gap by assigning to each year's average certified emissions, gap values from the literature considering the proportion of gasoline and diesel and the difference in the pollutant emissions legislation between the EU and Australia (Transport Energy/Emission Research, 2019; Smit, 2023). Morfeldt et al. (2021), analysing the effect on carbon footprint evolution from banning internal combustion engine vehicles (ICEVs) in Sweden, assumed average vehicle characteristics and CO₂ emissions of ICEVs, PHEVs, and BEVs for 2020 and 2030, and increased the certified values by a fixed percentage to account for the real-world conditions and adopted different average electric driving share of PHEVs in the two periods. Similarly, Hooftman et al. (2020), analysing scenarios for the EU passenger cars fleet decarbonisation up to 2050, assumed a progressive reduction of the average certified CO₂ emissions of ICEVs and PHEVs, and considered a real-world gap that progressively converged to 20% after 2030.

The methodological approach used in these studies incorporates vehicle datasets as input, usually only considering average fleet CO₂ emissions values per registration year or including limited vehicle characteristics. The real-world influence is added as a constant factor following existing literature, and assuming future developments. Such methodologies do not consider information for environmental conditions and vehicle utilization. Lately, BEV sales have been increasing worldwide, and the importance of ensuring similar performance under real-world conditions has been recognised. Electrification is the main sustainable solution that can be widely deployed, at the moment, given existing energy and fuels production infrastructure, market development and sustainability criteria. A 2024 thorough review of Life Cycle Emissions studies commissioned by the US Environmental Protection Agency (EPA) confirmed this observation (Eastern Research Group, 2023). Still, studies acknowledge the importance of clean energy production to decrease the environmental impact and mitigate climate change (Shafique and Luo, 2022; Zahoor et al., 2023). Thus, the introduction of holistic emissions quantification for BEVs and, in general, zero-emissions vehicles (ZEVs) should be expected soon worldwide, and evidence of the actual performance of ZEVs will be essential. According to Regulation (EU) 2019/631, the EC shall also assess the impacts of establishing minimum energy efficiency thresholds for new ZEVs placed in the EU.

The objective of the present study is to provide quantified evidence of the CO₂ standards' effectiveness, as a consequence of past decisions made, analyse the current fleet characteristics and potential, and discuss the shaping of future fleets based on recent regulatory developments. To address previous studies' limitations, the present study proposes an alternative method that integrates a detailed simulation framework to understand how the implementation of standards regulations is reflected under different scenarios. Compared to studies that perform qualitative analyses or studies that perform explanatory analyses using complex statistical models on available datasets, the framework follows a different approach to calculating the performance of passenger cars in selected certification protocols and on-road conditions. The advantage of this method is that parameters influencing specific operation or usage, for example, specific vehicle aerodynamics in highway driving, will be

⁶ FCAI, 2020: CO₂ Standard: Rules for Calculating Brand Targets and Assessing Brand Compliance (https://www.fcai.com.au/wp-content/uploads/2023/02/fcai_rules_for_calculating_co2_compliance.pdf).

reflected in the results. Together with the real-world framework and the parameters selected as inputs to it, the study simulates the fuel and energy consumption and CO₂ emissions of all powertrains, including BEVs, in annual usage. Thus, instead of attempting to explain the real-world performance using available parameters, the framework independently calculates the real-world CO₂ emissions according to the fleet characteristics and powertrain mix that the CO₂ regulations mandate. Given the inherent differences between the EU and the Australian fleets and the recently voted CO₂ standards in Australia, starting in 2025, the study applies the framework to both regions to provide insights from two different starting points and two paths with the same goal, to meet carbon neutrality. This study intends to give valuable insights to researchers and policymakers focusing on some of the biggest global markets that still don't have a solid road vehicle decarbonisation regulation.

2. Data and methods

2.1. Overview

Car registrations, together with vehicle characteristics, have been collected for both the EU and Australia to analyse the characteristics of the fleets registered in past years in both regions, as an outcome of the applied regulatory schemas. A comprehensive evaluation of the fleet characteristics registered in 2020 was performed and compared against the provisions of the 2020 targets. For a more recent year, 2021, vehicle emission simulations were performed to analyse the emissions performance following the certification protocols, as well as their differences to real-world conditions. Based on the latter, future projections on the need for electrification and the outcome are presented. Fig. 1 illustrates a schematic diagram of the study workflow.

2.2. Fleet historical data

The European Environment Agency (EEA) monitoring datasets⁷ were used for the annual registrations of new passenger cars in the EU: 2013 was selected as an early stage of the CO₂ targets implementation, the year 2018 as a mature stage, and the year 2021 as a WLTP-based certification and registration year, with stricter targets applying. The EEA monitoring data, besides the certified CO₂ emissions, provide a few technical specifications for each individual vehicle registered, such as powertrain type, vehicle mass, fuel type, engine capacity, and engine power. These data are not enough for performing detailed vehicle simulations which will be needed to understand the real-world performance. The 2021 monitoring data were combined with other online sources to expand the information for each vehicle. 257 representative make/model families for the EU, which correspond to 80% of the total sales, were produced for 2021.

For Australia, the construction of the registrations was based fully on online sources because there was no access to detailed registration data from FCAI. For the registrations of 2021, 118 different make/model families with the most sold conventional and electrified vehicle models were obtained for Australia, with their sales corresponding to 90% of the total sales of passenger vehicles in 2021. For the 2018 registrations, information from the 2018 Australian National Transport Commission (NTC) annual LDV CO₂ report⁸ was combined with online sources to reconstruct the same year's registrations.

The data collected for the 257 and 118 representative vehicle models

of 2021 registrations for the EU and Australia, respectively, are presented in Table 1. More details on the sources and the methodology for the selection of the representative models are presented in Appendix A.

2.3. Vehicle simulator

Vehicle simulations were performed in the reduced datasets of passenger cars 2021 registrations constructed for the EU and Australia. The selected simulator, PyCSIS, is a longitudinal dynamics physical model that simulates energy consumption and CO₂ emissions using a speed-time profile as input. It is a light version of CO₂MPAS (Fontaras et al., 2018), suitable for fleet simulations, since only a few vehicle characteristics (listed in Table 1) are needed. More details of the model and its accuracy can be found in previous studies (Tsiakmakis et al., 2017; Komnos et al., 2022).

PyCSIS uses pre-calibrated models extracted from a database of modern vehicles simulated with CO₂MPAS. When simulating a car with specific characteristics, the model uses the closest vehicle from this database. PyCSIS models all vehicle components, such as the drivetrain and the electric system, and uses a thermal model to simulate the vehicle's cold-start operation. For regular HEVs that do not integrate off-vehicle charging, also called Non-Off Vehicle Charging (NOVC-HEV), parameters have been used from the CO₂MPAS integrated hybrid model. For BEVs, a power-based model has been implemented and validated via experiments in previous study (Komnos et al., 2022) showing an acceptable accuracy of $\pm 10\%$ under real-world conditions. For Off Vehicle Charging hybrids (OVC-HEVs or PHEVs), the PyCSIS simulation is performed twice: once as a NOVC-HEV and once as a BEV. Regarding the vehicle resistances, PyCSIS uses predefined air drag coefficients for each body style, and rolling resistance coefficients, which were calibrated using a comprehensive dataset of EU WLTP-based type approval information (Chatzipanagi et al., 2022). More details can be found in previous study (Komnos et al., 2021).

Regarding the simulation of certified CO₂ emissions, PyCSIS is predominantly focused on WLTP. Besides the differences in the speed-time profile used in the NEDC compared to WLTP, NEDC allows several flexibilities during the testing, such as increased cell temperature, which limits the impact of engine cold start, vehicle preconditioning, lower vehicle inertia, and wider tolerances in deviating from the target speed. All these differences have been covered in the simulations following the NEDC to WLTP translation methods set out in Regulation (EU) 2017/1153 (more details in Appendix B). For the OVC-HEVs, the calculation of the simulated declared CO₂ values follows the steps of the actual certification protocols with few simplifications. The OVC-HEVs declared CO₂ emissions are calculated by weighting the emissions from a test performed under charge-sustaining (CS) operation and the CO₂ emissions from a sequence of charge-depleting (CD) tests. The weighting is done using a utility factor (UF_{TA}), which provides the probability that the daily driving needs will be met in full electric mode. Thus, the UF_{TA} is vehicle specific, and for the present study it is calculated using the vehicles' certified electric ranges (WLTP or NEDC-based). More details on the calculation steps are presented in Appendix B. As a side note, with the current UF_{TA, WLTP} formula in the regulation, the electric range of the majority of OVC-HEVs results in a UF_{TA, WLTP} range of 0.75–0.9, meaning that most of the driving needs can be met fully electrically.

Emissions in real-world driving can vary from those measured in official tests due to a range of factors. The majority of the factors, listed in different sources (Fontaras et al., 2017; Fan et al., 2023), have been considered to the extent possible.

- Additional load was added on top of the vehicle's actual mass to account for average passenger occupancy in the real world.
- Auxiliary systems such as steering assist, infotainment, and lights have been considered by adding an average extra electric load.

⁷ Monitoring of CO₂ emissions from passenger cars Regulation (EU) 2019/631 (<https://www.eea.europa.eu/data-and-maps/data/co2-cars-emission-20>).

⁸ NTC, 2019: Carbon Dioxide Emissions Intensity for New Australian Light Vehicles 2018 (<https://www.ntc.gov.au/sites/default/files/assets/files/Carbon%20dioxide%20emissions%20intensity%20for%20new%20Australian%20light%20vehicles%202018.pdf>).

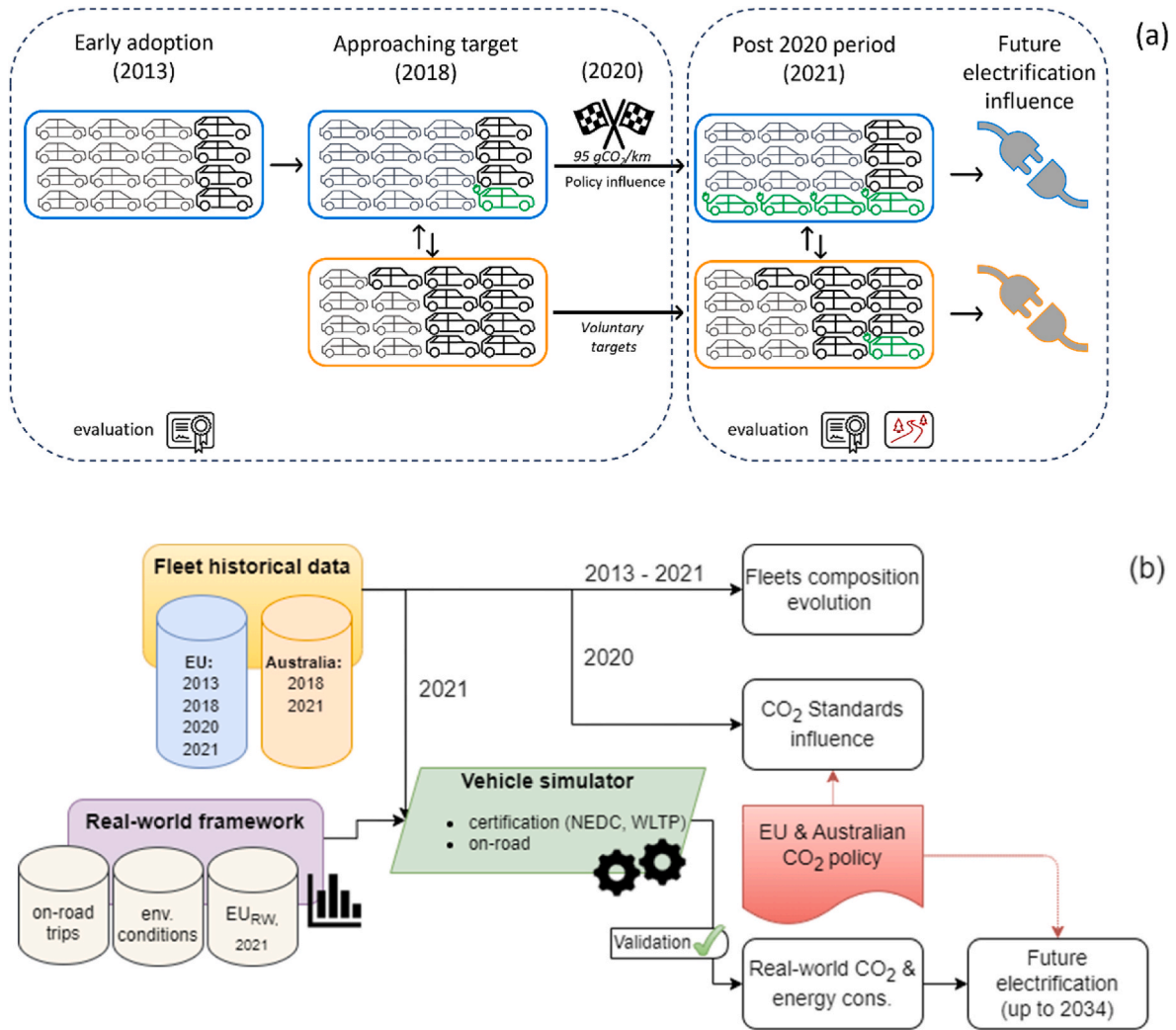


Fig. 1. (a) Study overview. Blue-coloured frames show the EU passenger new sales, and orange-coloured frames show the Australian passenger new sales. Differences in vehicle size are shown as well with small/medium cars (🚗) and large cars, SUVs and utes (🚙). Black-coloured vehicles correspond to conventional ICE vehicles, green-coloured to electrified vehicles. Fleets were evaluated against certification up to 2020 and after that, both for certification (📄) and real-world performance (🚗⚡). (b) Study workflow. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

- The effect of ambient temperatures on aerodynamics has been addressed by normalising the aerodynamic coefficients. Additionally, the impact of the cold-start is considered with a specific model.
- Heating, ventilation, and air conditioning (HVAC) are important energy consumers in real-world conditions (Komnos et al., 2022; Al-Wreikat et al., 2022), and they are modelled as a function of the ambient temperature.

The details of the above factors and their impact on fuel and energy consumption have been already analysed in a previous study (Komnos et al., 2022). Factors related to vehicle aging, such as tyre pressure and battery degradation, have not been considered due to their complexity and the lack of solid evidence in the literature. Additionally, a calculation procedure was developed to weigh the CS and the CD operation of an OVC-HEV (PHEV) via a UF_{RW} for real-world conditions. The detailed procedure and formulas are explained in Appendix B. Studies have shown that the UF_{RW} of OVC-HEVs is lower than the UF_{TA}, and the findings are discussed in the results section.

2.4. Real-world framework

This section describes the framework developed for replicating annual real-world driving usage together with the vehicle simulations.

2.4.1. General

A database containing 483 trip recordings of a campaign with an SUV performed in Italy in 2018 (Pavlovic et al., 2020) was used as a basis for the definition of real-world trips for both the EU and Australia. To reduce the computational cost of simulating all the trips recorded, a limited number of representative trips was only used. For the selection of representative trips, first, ten distance groups were derived. The distance of each group for trip distances up to 30 km was constrained to 5 km to capture a variety of traffic conditions and dynamics resulting from driving behaviour. The frequency of trips with distances above 30 km is lower, because the trips show more unified characteristics. Thus, it was decided to use a distance group of 50 km. Three trips were selected in each distance group, by applying a K-Mean classification in the trip distance versus average trip speed domain of each distance group. More details on the procedure of the selection of the 30 trips can be found in previous study (Komnos et al., 2022), while Appendix C provides further

Table 1

Data collected for the two passenger cars fleets registered in 2021. ICE: Internal combustion engine.

Powertrain	Parameters	Comments
Across all powertrains	Make and model	The models corresponding to the highest sales in a year were identified
	Powertrain	Pure ICE, HEV, PHEV, BEV
	Sales	Annual registration numbers
	Vehicle body	Sedan, hatchback, SUV, etc.
	Tare mass	the weight of an empty standard vehicle with all its fluids and 10 L of fuel in the tank
Pure ICE, HEV, PHEV	Tyre dimensions	From tyre code
	Body dimensions	Frontal area
	Engine capacity	–
	Engine max power	–
	Certified CO ₂ emissions	Only for validation
PHEV, BEV	Number of gears	–
	Motor max power	–
	Motor max torque	–
	Battery capacity	–
	Certified Electric range	For PHEVs
	Certified Electric consumption	Only for validation

insights into the representativeness versus the full real-world database. It should be noted that, although the study builds on the specific campaign in Italy, the method detailed below uses the velocity and elevation profiles for creating a generic real-world vehicle usage where the shares of urban/rural/highway driving are user inputs. In this way, the real-world driving is disentangled from the actual campaign. The workflow of the framework is presented in Fig. 2.

The inputs to the framework are three (Fig. 2). First, an annual distance shall be selected, which for the present study is derived from real-world data received for the EU fleet registered in 2021, as explained in section 2.4.3. Second, a share of urban/rural/highway driving. Although different vehicle types can be linked to different usage, a generic distance share of 40%, 20%, and 40% for urban, rural, and highway driving was selected. These values have been based on Eurostat⁹, a study on new mobility patterns in European cities (Armoogum et al., 2022), and monitoring data collected in real-world campaigns from the European Joint Research Centre (JRC) (Tansini, 2023). Based on the high urban population percentage in both regions, above 70% for both the EU¹⁰ and Australia¹¹, and the absence of further information, the same percentage of urban/rural/highway was assigned to both regions. The third input is the real-world CO₂ emissions map, created by interpolating the 30 trips simulated for each individual vehicle with PyCSIS, as explained in the vehicle simulator section.

The consequent calculations are presented in the respective box in Fig. 2. Examination of the characteristics of the speed profiles, both from the full dataset of the 483 trips, but also in the reduced dataset of the 30 trips, showed that most trips with distances below 10 km correspond to urban driving, while the majority of trips with distance above 75 km have a dominant share of highway driving. Trips with distances in the range between 10 and 75 km, were attributed to rural driving. For the collected trip distances of the three driving areas, average trip speeds are produced following the probability distributions of average speeds according to the trip distance (distributions coloured in red in Fig. 2).

⁹ Eurostat: Passenger mobility statistics. Data extracted in November 2021 (https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Passenger_mobility_statistics).

¹⁰ Eurostat, 2020: Urban and rural living in the EU (<https://ec.europa.eu/eurostat/web/products-eurostat-news/-/edn-20200207-1>).

¹¹ Australia Urban Population 1960–2024 (<https://www.macrotrends.net/global-metrics/countries/AUS/australia/urban-population>).

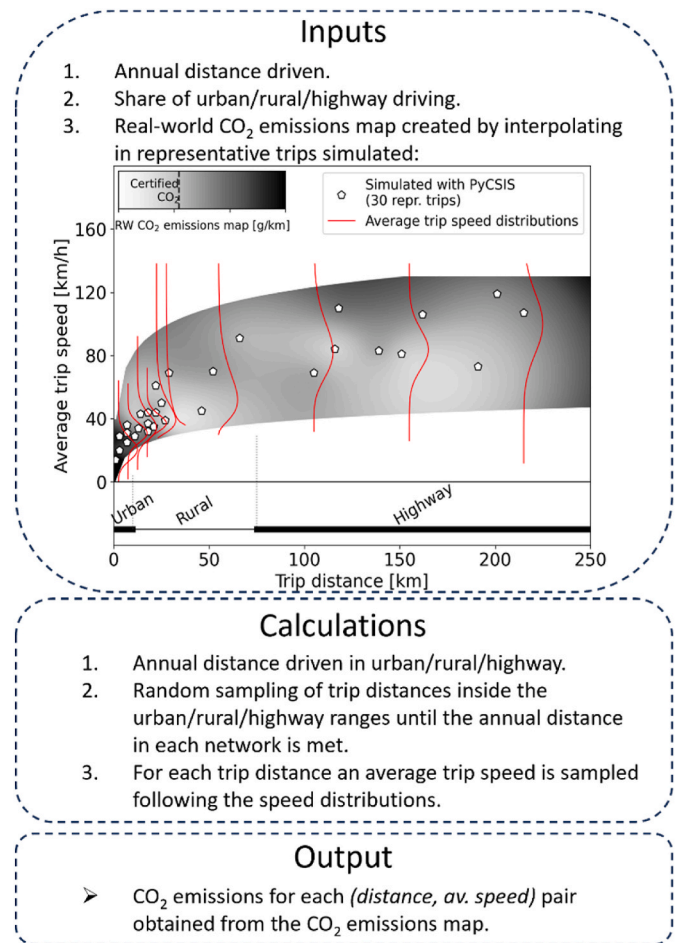


Fig. 2. Real-world framework steps followed for each individual vehicle model from the fleet.

Evidence (Pevce et al., 2019) that BEVs are driven more consciously to increase their range was addressed by decreasing by 10% the average trip speeds produced by the probability distributions. For each (trip distance, trip average speed) pair, a real-world CO₂ emission value or energy consumption value is calculated from the surface fitted to the simulated real-world values (points coloured white in Fig. 2). The annual distance-weighted CO₂ emissions or electric energy consumption is calculated using equation (1).

$$Consumption_{RW,annual} = \frac{\sum consumption_{trip} \cdot Distance_{trip}}{\sum Distance_{trip}} \quad (1)$$

2.4.2. Environmental conditions

Previous study (Komnos et al., 2022), assessing the influence of the ambient temperatures, demonstrated that considering all parameters, examining only the average annual ambient temperature, the real-world consumption can be underestimated as vehicle operation in ambient conditions between 14 and 20 °C has a reduced cold-start and HVAC effect. Additionally, the same study suggested that these parameters show a linear increase in CO₂ emissions and energy consumption.

An analysis of the ambient temperatures for the EU and Australia revealed substantial differences. For the EU, the average annual temperature was considered 14 °C, as described in the EU certification regulation (European Union, 2017). For Australia, vehicle kilometres travelled (VKT) weighted value of 18 °C was computed using data from another study (Smit, 2014). To account for ambient temperature variability across the year, the 2021 representative vehicles from the two regions were simulated in additional temperatures, extracted from

detailed analysis (more in [Appendix E](#)): ambient temperatures of -5 °C and 23 °C for the EU, and temperatures of 7 °C and 29 °C for Australia.

2.4.3. Real-world monitoring data for validation

The vehicle simulator together with the real-world framework was validated against real-world CO₂ emissions data reported for the 2021 EU-registered fleet, and presented in ([European Commission, 2024](#)). The 2021 real-world dataset, hereafter denoted as the EU_{RW,2021} dataset, reports the total fuel consumed and total distance travelled for about 620 thousand vehicles operating in the EU in 2021. The fuel consumption values are translated to CO₂ emission using standard values according to the fuel type of each vehicle, as presented in the first real-world report assessing the 2021 data ([European Commission, 2024](#)). More information about the origin of the data and the data processing is available in [Appendix C](#).

Equation (2) presents the calculation formula for the CO₂ emissions and energy consumption gap; the calculation was performed for each individual vehicle and then aggregated at the fleet level. This method differs from the one described in ([European Commission, 2024](#)), where the total real-world CO₂ emissions from a group of vehicles are summed and divided by the sum of certified CO₂ emissions of the same group of vehicles.

$$Gap_{individual\ vehicle} = \left(\frac{Consumption_{RW,annual}}{Consumption_{Certified}} - 1 \right) \cdot 100 \quad (2)$$

In the proposed real-world simulation framework presented in section 2.4.1, the annual distance is input for determining a vehicle's real-world consumption according to its body style. Average yearly distances per vehicle body style were derived from the EU_{RW,2021} dataset ([Table 2](#)).

The validation was limited to a sample of the dataset that correspond to six Member States with high representation in the 2021 monitoring dataset: Poland, Germany, France, Italy, Spain, and Portugal. For each selected Member State, the vehicles reported in the EU_{RW,2021} dataset correspond to more than 4% of the total registrations of passenger cars in 2021. Additionally, their annual average ambient temperatures span a wide range that will validate the impact of real-world conditions. For each Member State selected, a reduced dataset of representative model families was constructed with the same procedure explained in section 2.2. Additionally, each Member State's specific environmental conditions were considered to cover factors such as the driving patterns and fleet compositions (see [Appendix E](#)).

3. Results

3.1. Fleet-wide emissions

The 2018 Australian registrations constructed were validated against data from detailed studies ([Transport Energy/Emission Research, 2019](#); [Smit et al., 2024](#)) showing an acceptable qualitative and quantitative accuracy in terms of fleet-average weight, powertrain shares, and fleet-average CO₂ emissions (difference below 4 g CO₂/km compared to official information). [Fig. 3](#) shows that the 2018 registered European and Australian fleets have a 51 g CO₂/km (NEDC) difference. The differences can be attributed to a range of factors, including marketing efforts and consumer preferences linked to cultural, socio-economic, and environmental factors, but they are not investigated further in this paper but remain topics for future research. It is likely that the use of unambitious voluntary CO₂ targets in Australia has encouraged the sale of large and heavy passenger vehicles and disincentivized manufacturers from selling smaller and overall more efficient cars¹². [Fig. 3](#) also

compares the top-selling makes/models corresponding to 50% of each fleet registered in 2018. There is a clear distinction in engine size preferred in the two regions that highly correlates to CO₂ emissions. Australian vehicles can be larger and heavier with bigger engines, corresponding to higher energy use and emissions than the EU vehicles.

[Fig. 4](#) projects the evolution of CO₂ emissions, split by powertrains, versus mass for two reasons: (a) because mass is an important parameter in explaining the CO₂ emissions of a vehicle, and (b), the mass is a parameter used in the CO₂ targets as the parameter to allow market diversity. Comparing the EU registrations in 2013 to the ones of 2018, a year that the target of 95 g CO₂/km was approaching, the average fleet-wide CO₂ emissions of Pure ICE vehicles were reduced only by 5 g/km. At the same time, the sales of NOVC-HEVs increased from 0.7% to 1.5%, and a small sales share of OVC-HEVs (1%) and BEVs (1%) appeared. Until 2018, the share of Pure ICE vehicles in annual vehicle sales in the EU was higher than 90%. In 2021, the powertrain shares changed drastically: Pure ICE vehicles corresponded to only 60%, with zero and low-CO₂ emitting vehicles (ZLEVs) covering 20% of the fleet (the reasons for this change are analysed in the following section). The fleet-wide NEDC-based average CO₂ emissions of passenger vehicles sold in 2018 was 120 g/km. Using the average 2020 NEDC to WLTP ratio of 1.21 ([Suarez et al., 2023a](#)), the WLTP-based average of 114 g CO₂/km in 2021 would translate to an NEDC-based average of 94 g CO₂/km. This results in a 22% reduction in the average CO₂ emissions from 2018 to 2021 in the EU. Contrary to the EU, the penetration of electrified fuel-efficient passenger vehicles in Australia was slow. The Pure ICE vehicles still comprised 98% of the Australian fleet in 2018, with the rest being mainly NOVC-HEVs (1.7%). In 2021, NOVC-HEVs gained a higher percentage of sales of 10%, and OVC-HEVs and BEVs together reached 2%. [Fig. 4](#) shows that there was no visible change in the fleet-average CO₂ emissions (NEDC) per powertrain, but the uptake in NOVCs decreased the overall fleet's CO₂ footprint from 170 g/km to 161 g/km (5% reduction).

3.2. Influence of the CO₂ standards

Although most Pure ICE vehicles registered across different years in the EU were relatively small and efficient compared to Australia, their CO₂ values were not reduced to the extent necessary to meet the 2020 fleet-wide target of 95 g CO₂/km. This was achieved with the inclusion of ZLEVs (BEVs and PHEVs with CO₂ emissions below 50 g CO₂/km). To better understand the link between the individual Pure ICE vehicle's CO₂ emissions versus the fleet-wide target, the formula to calculate the manufacturer's target for 2020, according to their fleet average mass in running order (MRO) ([European Union, 2019](#)), M , presented in equation (3), is applied for each individual vehicle inserting their individual MRO, and then it is compared to the individual vehicle's certified CO₂ emissions.

$$EU\ CO_2\ Target_{passengers}^{2020-NEDC} = 95 + 0.0333 \cdot (M - 1,379.88) \quad (3)$$

[Table 3](#) shows the relevant average statistics of this calculation. [Table 3](#)-first row shows the difference from the 2020 target line considering only the Pure ICE average CO₂ emissions: in the fleet registered in 2013, the average difference from the 2020 target line was 33 g/km. This difference hardly changed in the following years. In 2019, a marginal increase in fleet-average CO₂ emissions, attributed to increased fleet-wide mass, regulation transition, manufacturers' sale strategies, did not affect manufacturers meeting their targets based on the 2015 fleet-wide target of 130 g/km¹³. The 2020 target could only be achieved by increasing the share of HEVs and BEVs as shown in [Table 3](#)

¹² TER, 2023: A Fuel Efficiency Standard for Light-Duty Vehicles in Australia (https://www.transport-e-research.com/_files/ugd/d0bd25_89c9da85af74c248ef7bbaf17d86297.pdf).

¹³ ICCT, 2020: CO₂ emissions from new passenger cars in Europe: Car manufacturers' performance in 2019 (<https://theicct.org/publication/co2-emissions-from-new-passenger-cars-in-europe-car-manufacturers-performance-in-2019/>).

Table 2Average annual distances per vehicle body style as derived from the EU_{RW,2021} dataset.

	convertible	coupé	hatchback	MPV	SUV/crossover	van	sedan	station wagon
Average annual mileage [km]	7433	7100	9138	14470	8723	14523	16488	15987

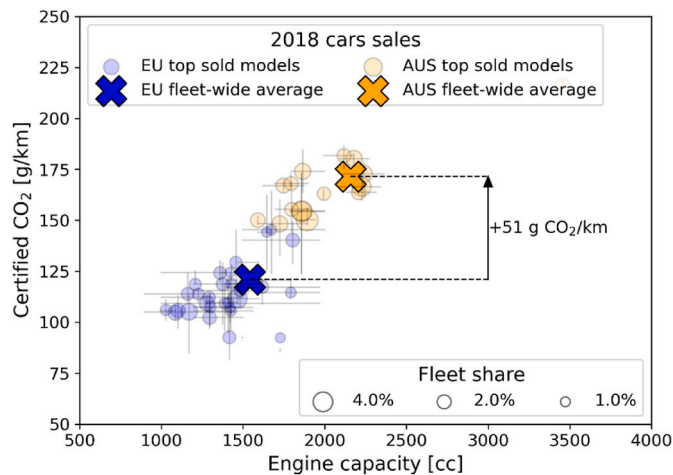


Fig. 3. NEDC-based fleet-average CO₂ emission rates for the 2018 top sales makes/models for the EU (blue) and Australia (AUS) (orange) versus engine capacity. Horizontal and vertical lines show the range (1st and 3rd quartiles) of engine capacity and CO₂ values, respectively. The size of the markers corresponds to each individual make/model percentage share in the total registrations. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

(second to the fourth row). It is evident that the registrations of electrified vehicles progressively increased in 2019 and 2020, with a clear drop in the fleet-wide average CO₂ emissions being observed only in 2020 (Table 3-last row). It should be noted that the results presented in Table 3 were calculated without the inclusion of regulatory provisions to support (pools of) manufacturers in meeting their targets. Such provisions are the super credits – vehicles with emissions below 50 g/km counted as more than 1 in the period 2020–2022 fleet average CO₂ emissions calculation, phase-in – exclusion in 2020 of a percentage of the fleet with the highest CO₂ emissions, and eco-innovations – emission savings for vehicles using eco-innovative technologies are accounted for.

The registration year 2020 is further analysed to better understand the importance of ZLEVs in meeting the CO₂ targets. Notwithstanding it is NEDC-based, assessing this year is important because it involved a strong fleet-wide CO₂ reduction from 130 g/km in 2015 to 95 g/km. The next fleet-wide (WLTP-based) CO₂ target line is foreseen for 2025. The period 2021–2024 serves as a transitional to WLTP certification, where the manufacturers' 2020 NEDC-based targets are renormalised for their WLTP to NEDC ratio in their 2020 registered fleet. This time, all the regulatory provisions are considered in the calculations. Additionally to the provisions explained above, manufacturers can form pools to jointly achieve the targets of each calendar year (European Union, 2019). As a final remark, specific derogation targets are set for manufacturers, with sales up to 300,000, by applying for derogations. The non-derogated manufacturers' pool performance results are presented in Fig. 5-blue dots. Only one pool failed to meet their targets (higher by 1 g CO₂/km).

To understand the importance of low-emitting vehicles entered in 2020 in meeting the 2020 targets, a portion of these vehicles sold in 2020 in the EU registration dataset was replaced with HEVs. In more detail, one-third of the low-emitting vehicles per manufacturer pool was modified to be represented as NOVC-HEV. As a result, the BEVs fleet-wide representation was reduced from 6.2% to 4.2%, and OVC-HEVs (<50 g CO₂/km) from 4.7% to 3.2%. Below, the two scenarios explored for the newly introduced NOVC-HEVs.

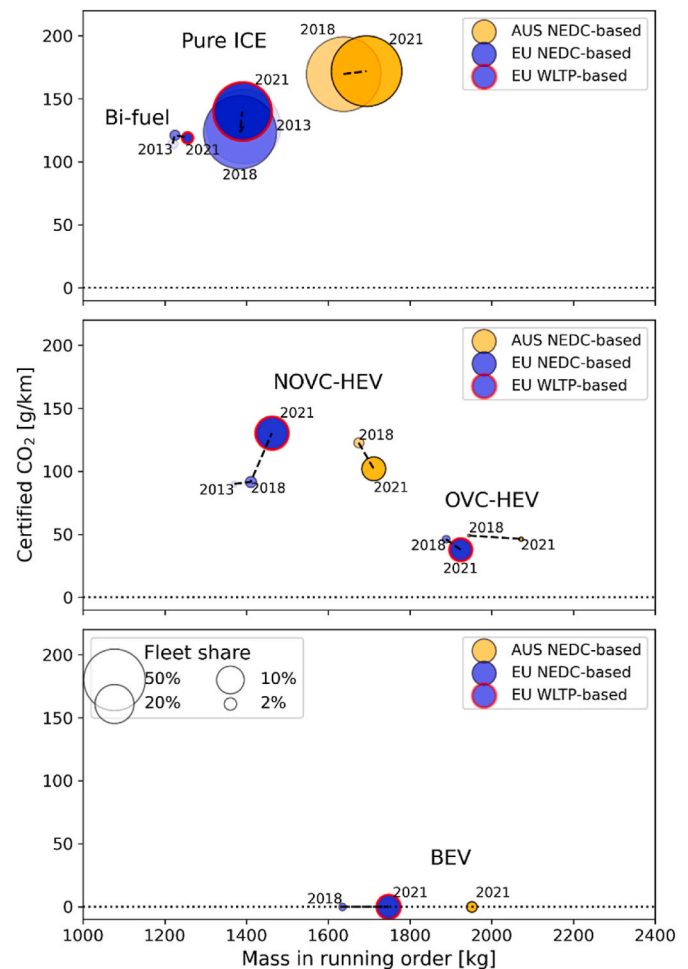


Fig. 4. Fleet average mass versus fleet average CO₂ for the EU (blue) and Australia (orange). Top figure: Pure ICE and Bi-fuel vehicles (only for the EU). Center figure: NOVC and OVC-HEVs. Bottom figure: BEVs. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

1. Replacing the CO₂ emission value with 109.5 g CO₂/km (i.e., the average NEDC-based CO₂ emissions of the registered NOVC-HEVs in 2020);

Table 3

Pure ICE difference from the 2020 target line, percentage shares of electrified vehicles in the NEDC-based assessment period, and fleet-wide CO₂ emissions calculated without the inclusion of regulatory provisions such as super credits, phase-in.

	Registration year			
	2013	2018	2019	2020
Pure ICE average CO ₂ difference from the 2020 target [g/km]	33	29	32	25
Share of NOVC-HEVs [%]	0.7	2.2	4.5	12.3
Share of OVC-HEVs [%]	0	1	1.2	5.1
Share of BEVs [%]	0	1	2.2	6.2
Fleet average CO ₂ [g/km]	127	121	122	108

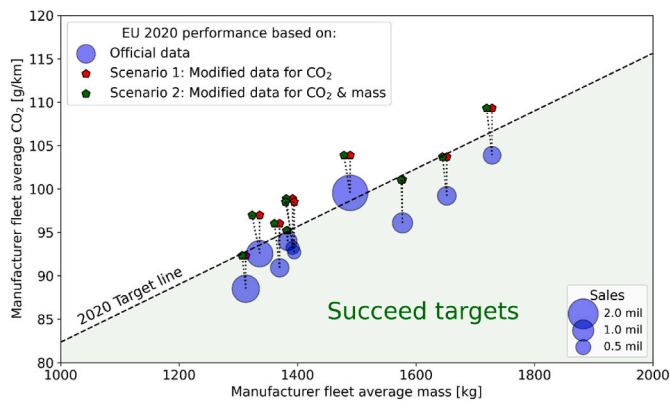


Fig. 5. EU 2020 CO₂ performance of the pools of manufacturers. In blue dots (their shape indicated the sales numbers) the eventual performance of the pools for 2020. Red and green dots show pools performance in the two scenarios. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

- Replacing the CO₂ emission value with 109.5 g CO₂/km and modify their mass to account for the difference in average mass with BEVs and OVC-HEVs. A decrease of 120 kg for the BEVs replaced by NOVC-HEVs was calculated, while for the replacement of OVC-HEVs, a 350 kg decrease was calculated.

With these adaptations, the sales-weighted CO₂ emissions from the whole fleet increased by 4 g CO₂/km. The results of the two scenarios are presented in Fig. 5, in red and green bullets, respectively. Seven out of the ten pools of manufacturers would fail to meet their target set in the two scenarios that reduce the ZLEVs. In the years following 2020, the popularity of electrified vehicles increased: in 2021, NOVC-HEVs constituted 19% of the registrations, OVC-HEVs 9%, and EVs 10%. In the provisional data for 2022, NOVC-HEVs covered 23% of the fleet, OVC-HEVs 9%, and EVs 13%.

The FCAI publishes the annual emissions performance of the manufacturers' newly registered fleets in Australia since 2020¹⁴. The agreement with its voluntary CO₂ Standard was assessed for two separate categories having different targets, MA (passenger cars and light SUVs) and MC + NA (heavy SUVs and light commercial vehicles). In the period 2020–2021, the overall performance of the MA category was well below the target, and the MC + NA category systematically failed to meet theirs. The voluntary Standard had certain limitations. First, the increased sales of heavy passenger cars (heavy SUVs and utes), and their corresponding influence on the total CO₂ emissions of passenger cars were hidden in the increased targets for light-commercial vehicles. In the three years of the voluntary targets (2020–2022), more than one-third of light-duty vehicles sold were allocated to the MC + NA category with a 50 g CO₂/km higher target as compared with the MA category. The second limitation comes from the lack of consequences if a manufacturer failed to meet the voluntary targets set. Australia will start applying excess emissions with the fleets registered since 2025 (Australian Government, 2024).

3.3. Simulation framework validation

To this point, the study focused on the evolution of vehicle registrations in the two regions until 2020. The next sections focus on the situation as seen starting from 2021 registrations with the introduction of the WLTP in the EU and real-world CO₂ emissions and energy

consumption. The latter is approached with simulations. The present chapter focuses on validating the simulation framework.

3.3.1. 2021 sampled car registration validation

Fig. 6 presents the sales-weighted distributions CO₂ emissions from the 257 models sampled for the EU and the 118 vehicles sampled for Australia for the fleets registered in 2021. For the EU fleet, the WLTP-based CO₂ emission rates from the reduced dataset are overlapped with data for the entire 9.7 million registrations (Fig. 6-left). The distributions and the fleet average CO₂ values from both datasets correspond well. Additionally, the reduced dataset averages to 113 g CO₂/km, 1 g CO₂/km lower than the full dataset.¹⁵ In Fig. 6-right, the 2021 Australian passenger vehicle fleet-average CO₂ emission rate, estimated using the mean CO₂ value of the sampled individual vehicle make/models, was calculated to be 161 g CO₂/km. This compares well with a recent TER/ICCT study (Smit et al., 2024), which estimated a fleet average value of 159 g CO₂/km, and the 160.6 g/km calculated by combining NTC/FCAI^{16,17} information for MA cars and SUVs in the MC + NA category.

3.3.2. Simulated CO₂ and electric consumption validation on certification protocols

Fig. 7 shows the correspondence of the simulated certified values of the 257 models sampled for the EU and the 118 vehicles sampled for Australia. Vehicles with ICE, ICEVs, for both regions, EU based on WLTP and Australian based on NEDC, show a high level of agreement with a coefficient of determination (R^2) above 0.84 for all powertrain technologies combined. BEVs have a lower level of agreement ($R^2 = 0.47$ for the EU and 0.33 for Australia). The weak correlation for Australia is less relevant due to the small sample size ($n = 6$). These results show an acceptable level of agreement for the purposes of this study. The divergence is mainly explained by the calculated vehicles' resistances and the over-declaration that was assumed to be constant (as explained in section 3.3). Considering the sales numbers, the simulated 2021 EU fleet-wide average CO₂ emissions were 114 g/km (WLTP-based), and the simulated 2021 Australian fleet-wide average CO₂ emissions were 158 g/km (NEDC-based). Compared to the official values presented in Fig. 6, there is less than 1 g/km difference.

3.3.3. Real-world simulation framework validation of CO₂ emissions

In Fig. 8, the gap results, as defined in equation (2), between the certified values versus the real-world consumption measured in the EU_{RW,2021} dataset, is shown as well as the simulated gap for each Member State and powertrain. For the Pure ICE vehicles (Fig. 8-left), the simulated results show a trend with the annual average ambient temperature. Poland has a higher gap than Germany, which has, on average, warmer weather. The EU_{RW,2021} dataset also shows a higher gap for Spain and Portugal, which can be attributed to a warmer climate and, therefore, higher use of air-conditioning on warm days. Although the warm days were considered in the simulations, the observed differences can potentially be explained by several factors that were not included in the present study, for instance, more dynamic driving, fuel properties, hilly or mountainous terrain, altitude effects, etc. For the NOVC-HEVs (Fig. 8-center), a difference of 5% between the values reported and

¹⁵ Monitoring of CO₂ emissions from passenger cars Regulation (EU) 2019/631 (<https://www.eea.europa.eu/data-and-maps/data/co2-car-emission-20>).

¹⁶ NTC, 2021: Carbon Dioxide Emissions Intensity for New Australian Light Vehicles 2020 (<https://www.ntc.gov.au/sites/default/files/assets/files/Carbon%20dioxide%20emissions%20intensity%20for%20new%20Australian%20light%20vehicles%202020.pdf>).

¹⁷ NTC, 2022: Carbon Dioxide Emissions Intensity for New Australian Light Vehicles 2021 (<https://www.ntc.gov.au/sites/default/files/assets/files/Carbon%20Dioxide%20Emissions%20Intensity%20for%20New%20Australian%20Light%20Vehicles%202021.pdf>).

¹⁴ FCAI, 2022: Monitoring CO₂ emissions from passenger cars and light commercial vehicles in 2021 (<https://www.fc.ai.com.au/monitoring-co2-emissions-from-passenger-cars-and-light-commercial-vehicles-in-2021/>).

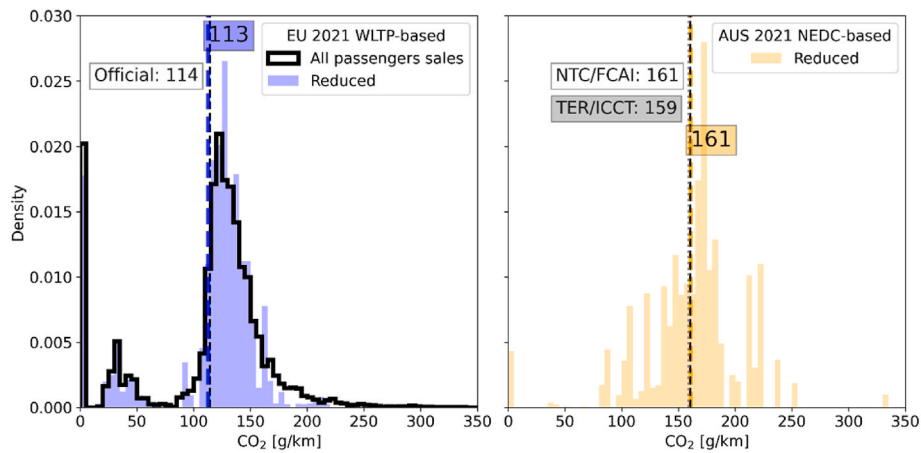


Fig. 6. The reduced passenger vehicle fleet CO₂ distributions for the EU (left) and Australia (AUS) (right). The vertical dashed lines correspond to the simulation average and the averages found in the literature.

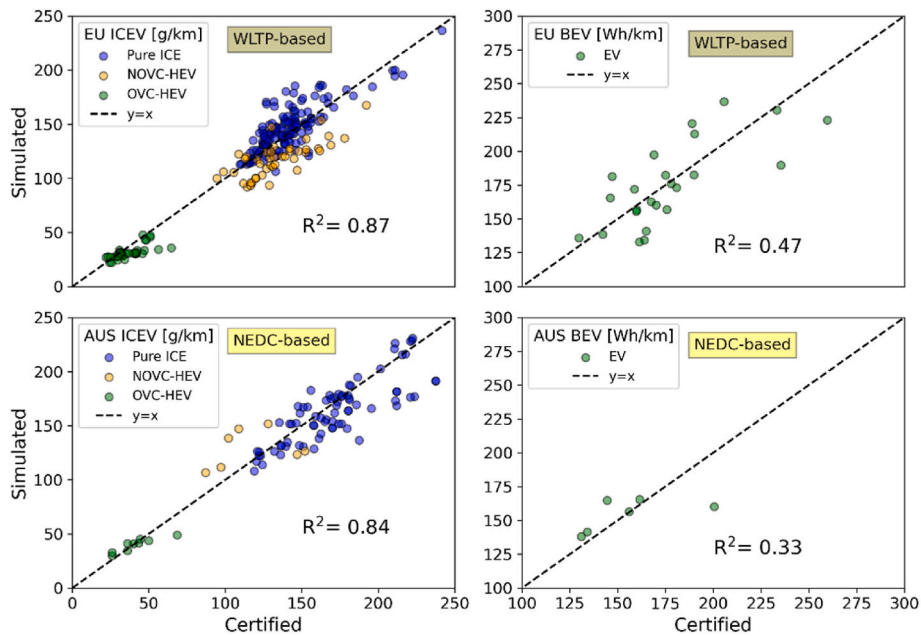


Fig. 7. Vehicle simulator validation using certification values for 2021 registered vehicles. The figures on the top are for the EU, and the figures on the bottom are for Australia (AUS).

the simulated is found. This limitation in the NOVC-HEVs simulation is solved later by increasing the real-world simulated consumption by 5%, only for that powertrain. For the OVC-HEVs (PHEVs) (Fig. 8-right), different UF_{RW} were tested, with the best agreement being $UF_{RW} = 0.25$.

The low UF_{RW} of PHEVs, calculated from the comparison to the real-world data, aligns well with evidence from multiple sources pointing to the different charging habits in corporate and private usage (Plötz et al., 2020; Plötz and Jöhrens, 2021), and the energy-based UF_{RW} , calculated by assessing the $EU_{RW,2021}$ dataset (European Commission, 2024). The $EU_{RW,2021}$ dataset does not distinguish between private and corporate cars in the reported consumption values.

As mentioned in the methodology details, the annual distance share of urban/rural/highway is an important input to the real-world framework. Driven by literature and real-world monitoring campaigns, the selected distance shares of 40%/20%/40% showed good agreement with the $EU_{RW,2021}$ data. The simulated real-world CO₂ emissions and the respective gap show sensitivity to this input. For example, the simulation results show that the OVC-HEVs would have a lower combined consumption by 15%, when the dominant use is urban conditions.

3.4. Real-world CO₂ emissions and energy consumption

Building on the simulation framework, this chapter stresses the CO₂ emissions and energy consumption as realised under real-world conditions. This analysis is performed for the registration year 2021 using the vehicle simulator and the real-world framework explained in the methodology and validated in chapter 3.3.

The sales-weighted average CO₂ emission values simulated for the vehicles registered in 2021 in both regions are presented per vehicle powertrain in Table 4. For the 2021 EU fleet, the simulated certified values are all close to the certified ones of the registered fleet (shown in parenthesis), and EU real-world values for Pure ICE and NOVC-HEVs are in the same order as reported for the first year of the real-world monitoring. The BEVs show an annual average gap of 27%, which is similar to fossil-fuelled passenger vehicles (17–21%). The OVC-HEVs were simulated considering a low $UF_{RW} = 0.25$, as calculated in section 3.3 by comparing to the $EU_{RW,2021}$ dataset. The resulting OVC-HEVs real-world CO₂ emissions are similar to small ICEVs CO₂ emissions. This is an important outcome for the potential of PHEVs. Even if their potential is

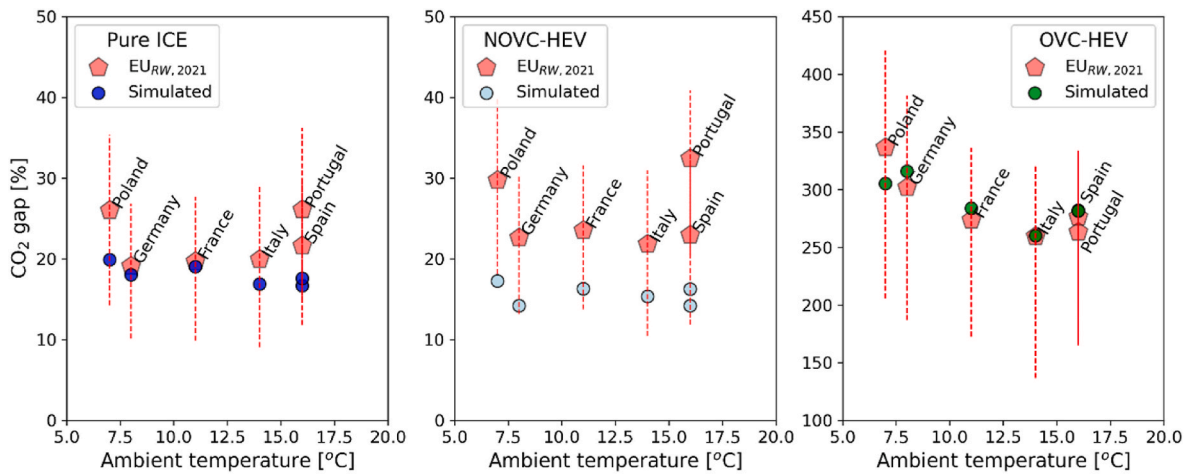


Fig. 8. Real-world framework validation. Simulation versus results from the EU_{RW, 2021} dataset. The vertical dashed lines show the CO₂ gap range (1st to 3rd quartiles).

Table 4

Simulation results for the EU and Australian (AUS) fleet samples in 2021. In parentheses, the information for the official average values is presented. The values in the parenthesis correspond to the fleet average values officially reported.

		ICEV ^a petrol	ICEV ^a diesel	PHEV petrol	PHEV diesel	BEV
EU	Sales	4,663,107	1,766,951	612,780	51,027	690,783
	Certified values (WLTP) ^b	133 (135)	139 (145)	32 (38)	36 (37)	172 (167)
	Real-world values ^b	161	163	123	128	216
	Gap [%]	21	17	299	278	27
AUS	Sales	518,945	92,900	962	–	13,020
	Certified values (NEDC) ^b	159	172	42		142
	Real-world values ^b	205	231	143		238
	Gap [%]	30	35	242		68

^a ICEV: Pure ICE and NOVC-HEVs together.

^b Certified and real-world values: For ICEVs and PHEVs the CO₂ emissions are presented in g/km. The electric consumption is presented for BEVs in Wh/km.

not explored in its full capacity, i.e., electric mode, their real-world average CO₂ emissions eventually drop the fleet's real-world footprint.

As shown earlier, the Australian fleet has higher certified CO₂ emissions. Table 4 shows that, having an NEDC-based certification, the real-world gap from the certified values is considerably higher than the EU. This gap is intensified for the BEVs, 68%, which can be partially explained by the intense use of HVAC auxiliaries for cooling due to higher temperatures during the summer period.

As mentioned in the methodology section, the annual distance shares were based on average values reported from different sources to be 40%, 20%, 40% for urban, rural, and highway conditions. However, it is logical that not everyone has similar habits and needs to fulfill. Additionally, the driving patterns are regularly imposed by the terrain (e.g., urban residency). The operational agility of the simulation framework serves to analyse the impact of different driving patterns, and Table 5 shows the results in terms of the percentage gap from official values. This analysis is significant, especially for BEVs whose usage might be influenced by their lower range compared to other powertrains and charging stations that are not equally distributed to urban, rural, and highway areas. Table 5 presents the results of the sensitivity analysis for

Table 5

Sensitivity analysis of the driving distance shares in the simulated real-world gap from the certified fleet average values. The driving shares are input to the simulation framework. The analysis is performed for both the EU and Australian (AUS) 2021 fleets. U: Urban; R: Rural; H: Highway.

	Pure ICE		NOVC-HEV		BEV	
U/R/H share [%]	EU	AUS	EU	AUS	EU	AUS
Base scenario (40/20/40)	19	30	25	34	27	68
Urban prevailing (80/20/0)	28	40	22	30	23	63
Rural prevailing (30, 70, 0)	14	25	18	26	20	59
Highway prevailing (10/20/70)	12	22	28	33	29	70

annual usage. The gap for the base case scenario for Pure ICE vehicles was calculated to 19% for the EU. However, in conditions where urban driving prevails, the gap reaches 28%, while in conditions where rural or highway conditions prevail, this gap is reduced to 12%. NOVC-HEVs and BEVs show a different picture: urban conditions favour their overall efficiency while prevailing highway conditions result in the highest consumption values. Similar trends were calculated from the simulations also for the Australian fleet. Such patterns are confirmed by the literature (Komnos et al., 2022; Mamarikas et al., 2020), and support the understanding of the real-world average values and their dispersion.

The BEVs' energy consumption gap from their certified values in the EU is estimated to be similar to that of conventional ICEVs, of the order of 25%, an increased value compared to what was previously reported (Komnos et al., 2022) at 5–10%. The reason is the different calculation methods employed, where in the past study higher weight was given to urban trips, assuming that these vehicles were primarily operated in urban conditions. The present study is based on the annual distance-weighted consumption, with the same driving shares as for other powertrain technologies allowing a direct comparison. Thus, highway driving, which is more energy-consuming for BEVs, has a considerable weight due to the higher distance accumulated compared to urban driving. Overall, the true gap of BEVs is expected to be somewhere between the findings of the two studies, a value that probably doesn't raise concern but should eventually also be monitored in the years to come to avoid further widening.

Fig. 9 shows the average tailpipe CO₂ emissions of the 2021 registrations as simulated under the certification protocols and real-world conditions. The CO₂ emissions are shown for their applicable test protocols. The emissions are presented against the procedure they were derived: MRO for the NEDC-based Australian average emissions, test mass, for the WLTP-based certified emissions, and a proxy of actual mass for the real-world vehicle utilization, as explained in the methodology

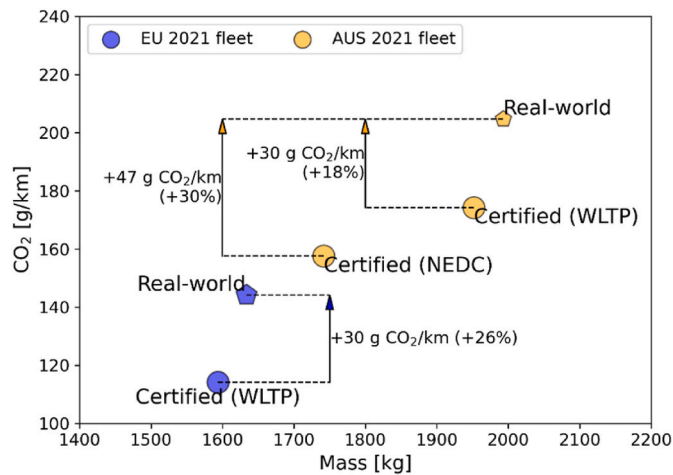


Fig. 9. Visual summary of certified and real-world fleet average CO₂ versus mass for the EU and Australia (AUS) in 2021.

section. For the Australian 2021 passenger car registrations (Fig. 9 – orange dots), the simulated fleet-average NEDC-based value of 157 g/km for 2021 was simulated to be 174 g/km in WLTP terms. This means that the EU and the Australian fleet have a difference of 60 g CO₂/km if they were compared based on the same protocol. The real-world CO₂ emissions were calculated to be 204 g CO₂/km for Australia, which is 30% higher than the NEDC-based certified CO₂, which is in line with an Australian PEMS study (Smit et al., 2022). Based on WLTP, the real-world CO₂ gap would be limited to 30 g CO₂/km, i.e., 18%, which is in line with other studies that reported a gap of up to 17% (Komnos et al., 2024). For the EU, the total CO₂ gap calculated was 30 g CO₂/km, i.e., 26% of the average WLTP-based CO₂ emissions (Fig. 9 – blue dots). The higher percentage gap in the EU is explained by the high percentage of OVC-HEVs in new vehicle sales in 2021 (9%), which, according to the UF_{FW} calculated (calculated to 0.25), has an average difference of 300%, which is a factor of four higher.

3.5. Future electrification influence

To gain more insight into the fleet average CO₂ emissions of the vehicles registered each year for the period up to 2035, the simulation results for 2021 fleets are used as a reference for the period 2021 to 2024, and projections for the next years are performed, following the regulatory schemas in the two regions. NOVC-HEVs show a progressive increase in their sales, reaching a share of 25% in the registration of new passenger cars in 2023 in the EU. Considering the last years' sales evolution, an average share of 35% for the period 2025–2029 and 40% for the period 2030–2034 was assumed for the EU. For Australia, the share of NOVC-HEVs in the 2023 new registrations¹⁸ was 8.1%. For 2025, the first year of mandatory targets applying, this share is assumed to be 15%, with an annual increase of 3% for the following years up to 2029. Regarding the sales of PHEVs, a stagnation in their sales was assumed for both regions, supported by the recently published sales information for 2022 and 2023. In the next step, following an iterative process, a percentage of the sales of pure ICE models is substituted by BEVs, or more generally ZEVs. The fleet-average CO₂ emission rate is then recalculated until it converges with the CO₂ target for a specific period.

The calculated forecast for ZEV sales for meeting the targets is simplified because specific provisions can't be implemented due to information that is not already available. The targets are assessed per

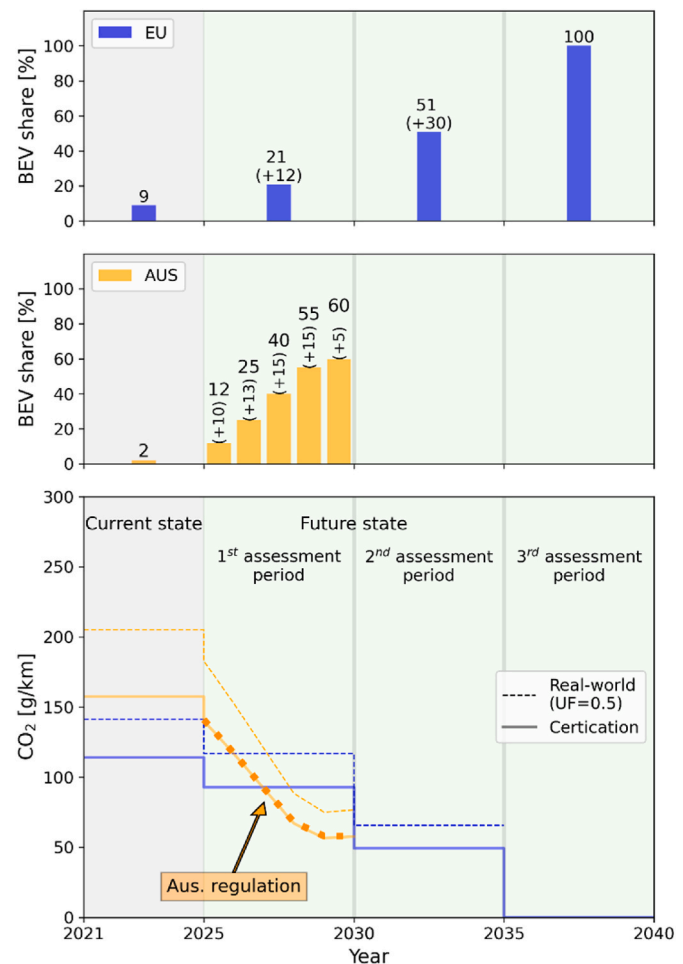


Fig. 10. BEV shares and influence on the fleet average CO₂ emission rates for EU and Australian (AUS) new vehicle sales based on the 2021 fleets simulation results. The two regions have different certification protocols. Hence, the certification lines are not directly comparable.

manufacturer (or pool of manufacturers in case of the EU), and not as a whole fleet. Thus, even if the targets set are met as a whole, specific manufacturers might fall above their CO₂ limits, facing fines. The ZLEV factor, considered in the EU for the period 2025–2029 is an example of the latter: manufacturers (or pools) with more than 25% of ZLEVs in their annual sales, will benefit by an increase in their CO₂ target. Additionally, the forecast does not consider the changes foreseen in the EU CO₂ certification calculation of the UF_{TA, WLTP} of PHEVs, scheduled for 2025 and 2027 (Regulation (EU) 2023/443). These changes will considerably lower the UF_{TA, WLTP} and it is expected to increase the declared CO₂ emissions of current PHEVs by two and three times if new developments to increase the electric range of PHEVs do not come forward. For what regards the real-world fleet CO₂ emissions of PHEVs, the study uses a UF_{FW} of 0.5, assuming that the electric driving share of PHEVs will increase compared to the findings of the 2021 EC report on the real-world CO₂ emissions gap (European Commission, 2024).

Fig. 10 shows the results of the projections. To meet the 2025 WLTP-based targets in the EU (93.4 g CO₂/km), 21% of the vehicles sold each year from 2025 to 2029 should be ZEVs. For the period 2030 to 2034, 51% of the cars sold each year should be ZEVs to meet the 2030 target (49.5 g CO₂/km). As a result, an increase of 12% in the sales of ZEVs appears to be necessary for the 2025–2029 period compared to the 2021 baseline. For the period 2030–2034, an extra 30% increase in ZEVs is required compared to the 2025 baseline. In contrast to the EU, where the target is implemented for longer-term periods, Australia has set annual

¹⁸ FCAI, 2024: Australia breaks all-time new vehicle sales in 2023 (<https://www.fcai.com.au/australia-breaks-all-time-new-vehicle-sales-in-2023/>).

targets for the period 2025 to 2029 (Parliament of Australia, 2024). To meet the NEDC-based CO₂ targets set for 2025 (141 g/km), ZEVs will need to cover 12% of sales of that same year. Similarly to the EU, the percentage of ZEVs will need to increase in the following years, reaching 60% in 2029 to meet the NEDC-based CO₂ target of 58 g/km. A report from the International Council on Clean Transportation¹⁹ (ICCT) reported similar results in a scenario where Australia aligns with World-class standards. The calculations show that the share of ZEVs needed for meeting the annual targets in the EU is smaller than in Australia. The reason is the fleet composition assumed: smaller and less-consuming conventional vehicles, with a higher percentage of NOVC-HEVs and OVC-HEVs in the EU compared to Australia. Fig. 10 additionally presents the real-world CO₂ tailpipe emissions of the fleets registered each year, accounting for the fleet compositions calculated. In the period 2030–2034, the calculated fleet composition of 40%, 9%, and 51% for NOVC-HEVs, OVC-HEVs, and ZEVs, respectively, would result in real-world fleet-average tailpipe CO₂ emissions of 66 g/km. For Australia, the calculated composition for 2029, 9%, 30%, 1%, and 60%, for conventional, NOVC-HEVs, OVC-HEVs, and ZEVs, would result in real-world fleet-average tailpipe CO₂ emissions of 77 g/km. For both regions, the growing share of ZEVs will have a consequent decrease in the absolute difference between real-world and certified fleet-average CO₂ emissions.

4. Discussion

Performing a practical scenario for the 2020 sales, the study showed that increased uptake of electrified vehicles was key for achieving the EU's 2020 CO₂ targets, without which manufacturers would have to pay fines. Instead, Australia's general lack of regulation has left this market with fewer model choices for alternative powertrains (Australian Government, 2024). Starting in 2025 and reaching 2030, the two regions have designed similar paths to progressively reduce passenger car emissions' impact on climate change.

In the following years, the powertrain composition of new fleets entering the markets is expected to rapidly diversify. In the present study, the existing regulatory schemas and knowledge were combined with the proposed simulation framework to perform forecasts of the actual performance of the fleets in the real world. Updates in the regulations will result in changes in vehicle characteristics and fleet compositions. For example, the update of the UF_{WLTP} of PHEVs, foreseen in the EU for 2025 and 2027, might result in PHEVs not considered as low-emitting at least with the current characteristics. As a result, manufacturers will either focus their sales on BEVs or will increase their PHEV batteries' capacity to increase their electric range. Australia is expected to replace NEDC with WLTP (Australian Government, 2024). The change in more realistic fuel and energy consumption values, and CO₂ emissions can result in the uptake of smaller and lighter vehicles in Australia. The examination of such changes will be important because fleets registered in the following years will be on the roads for the next 10–20 years. The framework proposed is suitable for detailed fleet analysis, able to accept information about the evolution of vehicle characteristics, offering an understanding of the impacts of replacing the certification protocols and the real-world performance of future passenger cars.

Fig. 11 shows that CO₂ targets are expressed as a CO₂ line and not a single cut-off threshold. Besides the target fleet-wide CO₂ value, two more variables are communicated: the fixed intercept and the slope of the line. The fixed intercept in both regions is selected to be a representative fleet-wide average mass. Updating this mass value every few years by calculating the fleet-average mass of the fleets of previous years ensures that the regulation accounts for new developments in the fleet's

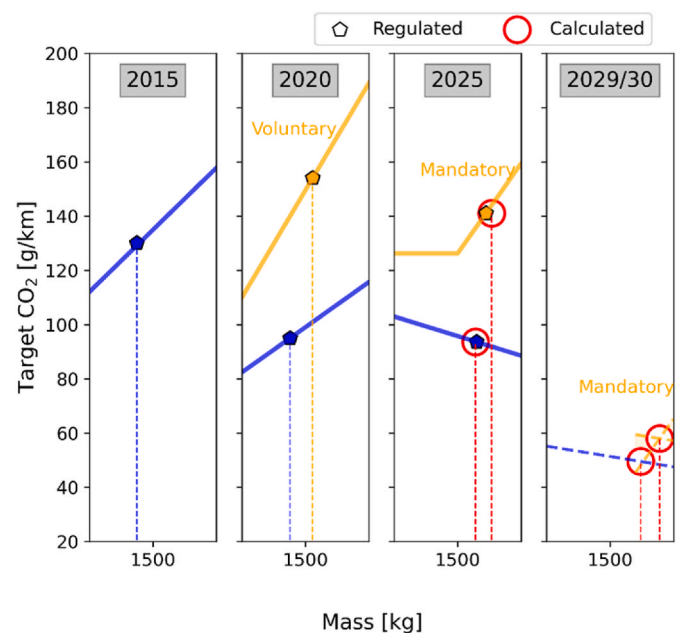


Fig. 11. CO₂ target lines versus mass. The EU mandatory targets are in blue. The voluntary targets and the proposal for 2025 onwards for Australia are in orange. It should be noted that the two regions have different certification protocols; hence, the target values are not directly comparable. The orange and blue dashed target lines in the period 2029 and 2030 indicate that the lines are not fully determined. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

compositions. As shown in Fig. 11, the projections in fleet compositions calculated similar mass values (red circles) to the regulated ones (pentagons) for 2025. For the period 2029 and 2030, an increase of 100 kg is calculated for the total passenger car registrations mass, compared to 2025. This is due to the strong uptake of ZEVs to support meeting the CO₂ targets set in both regions. Similar to the mass, the slope of the line is updated regularly to drive improvements in emissions and help shape a more efficient vehicle fleet. The calculated 2025 and 2030 EU passenger cars target lines have negative slopes as a result of the provisions of (EU) 2019/631, the high 2021 sales of BEVs and PHEVs that have zero and low CO₂ emissions, together with the higher masses of that powertrains compared to conventional vehicles (Suarez et al., 2023a). While the EU slopes introduce similar stringency and promote manufacturers with heavier fleets to introduce high shares of ZEVs, the considerably high positive slope proposed in Australia for 2025 could potentially lead to the introduction of heavier, less energy-efficient vehicles despite the introduction of CO₂ targets (Smit and Whitehead, 2024). Thus, analysis of the influence of the slope of the line and scenarios of what powertrain compositions will be needed per manufacturer to meet the targets set will be crucial.

Globally, the application of CO₂ emissions targets does not consider the emissions generated in the manufacturing of the vehicle, fuel or energy production, or at the end of a vehicle's life. Although the impact of ZEVs is calculated to be substantially lower than ICEVs (Australian Government, 2024; Bieker, 2021; Opetnik et al., 2024; Smit and Kennedy, 2022), promoting vehicles that consume less energy over their entire lifecycle is key for long-term sustainability and reaching carbon neutrality. Weiss et al. (2024) used an online EV database to study the correlation between certified and real-world energy electric energy consumption and vehicle attributes. While CO₂ emissions of Pure ICE and NOVC-HEVs are highly correlated with vehicle mass (resulting R² of 0.6), the mass does not explain well the electric energy consumption of EVs (R² of 0.3), neither does the battery capacity and drive range. Whether assessing the ZEVs inside the CO₂ Standard regulation or standing alone against ZEVs energy efficiency standards, simulation

¹⁹ ICCT, 2022: Fuel efficiency standards to decarbonize Australia's light-duty vehicles (<https://theicct.org/publication/pv-australia-co2-standards-dec22/>).

tools can provide indications of how each vehicle characteristic affects their energy consumption, both for the purpose of the certification and the real world. The lower correlation between vehicle attributes and energy efficiency of BEVs should increase with the addition of their lifecycle CO₂ footprint in their assessment; it is likely that such an addition will promote of smaller, lightweight, and reduced climate and material impact cars²⁰.

The present study has focused on the efficiency improvements of car fleets registered each year and how the CO₂ standards regulations influence them. While technologies carrying such improvements need time to be deployed and gain the needed share to make a difference in the whole fleet, demand-side solutions should be acknowledged for allowing immediate emissions and energy consumption reductions (Winkler et al., 2023; Creutzig et al., 2018). Thus, car-use reduction policies, such as supporting active travel (cycling and walking), improving public transit, and reducing the need for travel, will play an important, complementary role in an integrated policy mix (Axsen et al., 2020).

5. Conclusions

The study proposed a fleet simulation framework to assess the real-world consumption and the actual benefits of different powertrains. The framework was extensively validated, showing good agreement with the first year of real-world fuel consumption monitoring data in the EU. The comparison between the EU and Australian passenger cars fleets showed that the difference in the average certified tailpipe CO₂ emissions was estimated to be about 50 g/km in 2018, when both regions used the NEDC protocol. Simulations allowed the comparison of the two fleets registered in 2021 under the WLTP certification. Below some important findings are summarised:

- In the case of the EU a step-by-step analysis, following the provisions set out for the 2020 target of passenger cars showed that the sales uptake of ZLEVs in that year was crucial for meeting the target of 95 g/km. In a scenario where the ZLEVs sales were lower by one-third, seven out of the ten manufacturers pools would not meet their target. The fleet-wide CO₂ emissions, accounting for regulatory provisions such as the super credits, would be 100 g/km. The uptake of electrified vehicles in the EU market widened the difference in the certified fleet emissions from 50 to 60 g CO₂/km.
- In 2021, the average real-world emissions were estimated at about 143 g/km for the EU and 204 g/km for Australia. For the same year, the WLTP-based CO₂ emissions gap was calculated for the EU at 26% - or 30 g CO₂/km, including the effect of PHEVs. The NEDC-based gap in Australia was calculated at 30% - or 47 g CO₂/km. If Australia was using a WLTP-based certification, their vehicles would have more realistic certified values, lowering the gap to 18%.
- The sensitivity analysis results show that HEVs and BEVs perform better when driven in urban conditions, where the EU WLTP-based gap is calculated to be 22–23%, while in the base case scenario (40%, 20%, 40% for urban, rural, and highway driving) the gap was calculated to be around 25–27%. Instead, when driven mainly in

highway conditions, the calculated CO₂ gap was calculated to 28–29%.

- Using the 2021 registrations as a reference for the period until 2024, the study performed projections on the need for ZEVs for the periods 2025–2029 and 2030–2034, following the recent updates on the CO₂ regulatory frameworks of the EU and Australia. For the EU, about 51% ZEVs will be needed to meet the target of 49.5 g CO₂/km in 2030, while 60% ZEVs will be needed to meet the target of 58 g CO₂/km in Australia in 2029. The real-world fleet-average CO₂ emissions of those fleets were calculate at 66 g/km, and 77 g/km for the EU and Australia, respectively.

Overall, the methodology and simulation approach used in this study can be valuable to researchers and policymakers focusing on some of the biggest global markets that still don't have solid road vehicle decarbonisation regulations and help improve their emissions/energy consumption and fleet projection models.

As per the points for future improvement, biases in the predictions, for example, NOVC-HEVs simulations under real-world driving were manually corrected, and future work will focus on improving the simulation model. Additionally, since the HVAC auxiliaries are getting more attention for their real-world impact and manufacturers are trying to increase their efficiency, more detail in the model used will be added. There is an indication that BEVs are driven more cautiously due to the so-called range anxiety, which was partially addressed by decreasing by 10% the average trip speed. Future real-world campaigns will give more information about the driving habits and shares of urban/rural/highway driving that will improve the framework's accuracy and detail.

CRedit authorship contribution statement

Dimitrios Komnos: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Robin Smit:** Writing – review & editing, Validation, Data curation. **Leonidas Ntziachristos:** Writing – review & editing, Supervision, Methodology. **Georgios Fontaras:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. 2021 Fleets construction and data sources

The official datasets published yearly by the European Environmental Agency (EEA) include certified CO₂ values and a few technical vehicle characteristics. 9.7 million passenger cars were sold in the EU in the year 2021. For the present study, the vehicles were grouped by vehicle makes/models, powertrain, and fuel type, forming 800 model families. The highest sales families, covering 80% of the total passenger car registrations, were selected. The resulting 257 families are a compromise between computational cost and accuracy. The average WLTP-based certified CO₂ emissions

²⁰ Transport and Environment, 2024: Making EVs fit for the future (<https://www.transportenvironment.org/articles/making-evs-fit-for-the-future>).

were assigned to each family in the next step. Then, the data for the selected families were combined with other online sources to expand their vehicle characteristics^{21,22}. The reduced EEA dataset and the online data were combined by selecting the vehicle model's characteristics with the same powertrain and its certified WLTP-based CO₂ emissions, which have the minimum difference from the average value calculated for each vehicle family.

For Australia, there is no detailed open-source official data for the annual passenger car registrations. To overcome this limitation, online sources²³ presenting the sales of the 100 most popular makes/models yearly were used. These sales sources were expanded with online sources for the most popular hybrid electric vehicles (HEVs)²⁴, and BEVs²⁵. The sales information of the resulting 118 different makes/modes was combined with information from an official Australian portal, the Green Vehicle Guide (GVG) (DITRDCA, 2020), which provides a vehicle's NEDC-based CO₂ emissions. The GVG was also used to determine which vehicles were considered passenger cars. The body styles cab-chassis and utility were excluded, although utility vehicles (utes) are often used for personal travel and not as light-commercial vehicles²⁶. Similar to the procedure followed with the EU dataset, the mean CO₂ value for the different models was assigned to each vehicle model according to its powertrain: conventional, hybrid, or electric. The GVG information was not covering all the parameters needed for performing simulations (Table 1). Therefore, further details on the vehicle make-s/models were derived from online sources.²⁷ Finally, the Australian National Transport Commission (NTC) publishes aggregated sales and NEDC-based CO₂ emissions performance values per vehicle category and for the most sold vehicles each year. The 2020 and 2021 NTC reports were used to validate the passenger cars dataset created for this study. In 2021, 0.75 million passenger cars were sold in Australia, that corresponds to 7.5% of the passenger vehicles sold in the EU. The reduced dataset created for the 2021 passenger vehicles sold in Australia includes more than 90% of total sales.

A graphical representation of the combination of different sources for the 2021 EU and Australian registration datasets is presented in Fig. 12. The blue boxes are sources of information related to the EU fleet, while the orange boxes are associated with Australia.

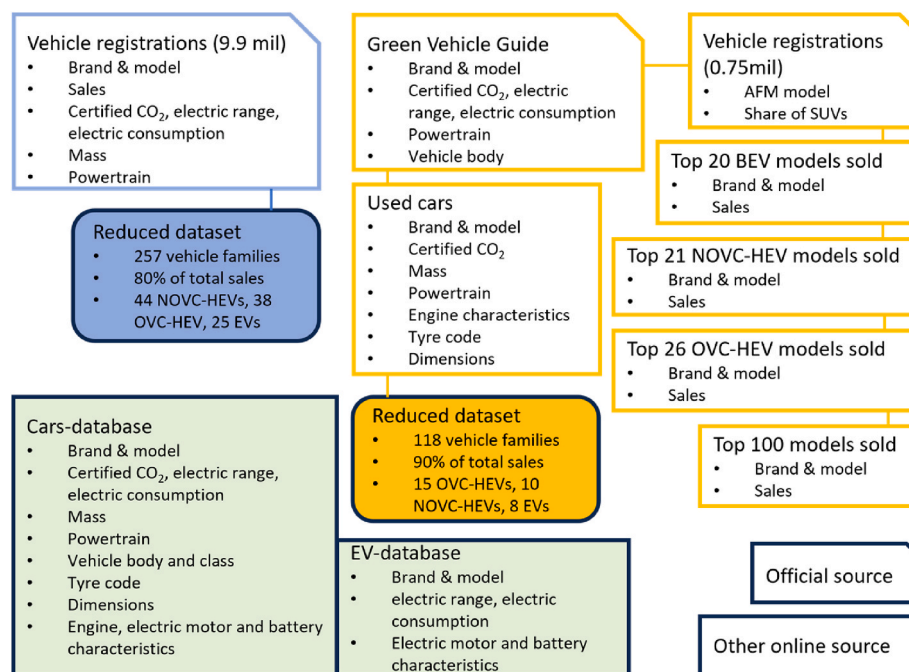


Fig. 12. Data sources combination for the EU (light blue) and Australia (orange) 2021 fleets recreation. Green boxes contain information used in both fleets' creation.

Appendix B. Vehicle simulator details

Differences in the certification protocols

Although the road load determination principle is the same, some critical elements differ between the NEDC and the WLTP. First, the vehicle mass considered in the NEDC is the MRO plus 25 kg (Regulation (EU) 2017/1153). For the WLTP, the optional equipment's mass and a percentage (15% for passenger cars) of the maximum permissible vehicle load are added to the MRO (European Union, 2017). The resulting WLTP test mass better represents real-world conditions. Other parameters affecting the road load coefficients are the tyre pressure and the tyre rolling resistance coefficient,

²¹ Car specs database (<https://www.cars-data.com/>).

²² EV database (<https://ev-database.org/>).

²³ Top 100 new cars sold in Australia in 2021: From Toyota HiLux and Ford Ranger to Mitsubishi's Pajero and Mirage (<https://www.carsguide.com.au/car-news/top-100-new-cars-sold-in-australia-in-2021-from-toyota-hilux-and-ford-ranger-to-mitsubishi>).

²⁴ Australia's best-selling hybrids and plug-in hybrids of 2021 (<https://www.drive.com.au/news/australias-best-selling-hybrids-and-plug-in-hybrids-of-2021/>).

²⁵ Australia's best-selling electric cars of 2021 – that aren't Teslas (<https://www.drive.com.au/news/australias-best-selling-electric-cars-of-2021-that-arent-teslas/>).

²⁶ Climate Council, 2023: Ute Beauty! The case for lower and zero emissions utes in Australia (<https://www.climatecouncil.org.au/resources/ute-beauty-case-for-lower-and-zero-emissions-utes-australia/>).

²⁷ Redbook Cars (<https://www.redbook.com.au/cars/>).

which were not prescribed in the NEDC. In contrast, for the WLTP, the exact specifications of the tyres are used for the test. On average, the above differences result in higher WLTP vehicle resistances, as outlined in Pavlovic et al. (2018). The NEDC-based vehicle resistances, in the form of road load coefficients, are estimated by converting the WLTP ones using the methodology described in the NEDC-WLTP correlation Regulation (EU) 2017/1153. The WLTP-based laboratory testing is simulated with the vehicle in an initial state of 23 °C, and the NEDC-based simulations were performed starting at 25 °C.

Contrary to NEDC, in WLTP, the manufacturer must declare a CO₂ value equal to or higher than measured to eliminate measurement uncertainty (Pavlovic et al., 2018). A previous study quantified the range of the over-declaration to 4–6% (Chatzipanagi et al., 2020). For the present study, an over-declaration of 4% was considered. Also, for electricity consumption, according to the WLTP regulation (Reick et al., 2021), the battery charging losses need to be accounted for. The charging losses of OVC-HEVs and BEVs were assumed to be 6%.

OVC-HEVs combined weighted CO₂ emissions

WLTP requires calculating the CD cycle range (R_{CDC}). R_{CDC} is the number of complete cycles driven in CD mode. Equation (B.1) is a good proxy of R_{CDC} via the electric range and the distance of the certification cycle (WLTP: 23.15 km). To host the regulation requirement to include the cycle where ICE turns on – and CO₂ is emitted, in equation (B.1), the ratio of electric range versus cycle distance is rounded up to the nearest integer. UF_{TA} in equation (B.2) is calculated from the certification function described in the Regulation (EU) 2017/1151. For the CD CO₂ emissions (CO_{2,CD,WLTP}), an average value was used for simplicity: CO_{2,CD,WLTP} = 11 g CO₂/km, which was derived by analysing type-approval data. As a result, the weighted WLTP CO₂, calculated as shown in equation (B.3), will range in very low values. The NEDC-weighted CO₂ emissions are calculated via equation (B.4) for the NEDC protocol. For simplicity, in the present study the CO_{2,CD,NEDC} = 0 g/km, which is safe to assume according to a study by Pavlovic et al. (2017).

$$R_{CDC} = \left\lceil \frac{\text{electric range}_{WLTP}}{\text{cycle distance}} \right\rceil \cdot \text{cycle distance} \quad (B.1)$$

$$UF_{TA} = f(R_{CDC}) \quad (B.2)$$

$$CO_{2,combined-weighted,WLTP} = CO_{2,CS,WLTP} \cdot (1 - UF_{TA}) + CO_{2,CD,WLTP} \cdot UF_{TA} \quad (B.3)$$

$$CO_{2,combined-weighted,NEDC} = \frac{CO_{2,CS,NEDC} \cdot 25 + CO_{2,CD,NEDC} \cdot \text{electric range}}{25 + \text{electric range}_{NEDC}} \quad (B.4)$$

Real-world driving considerations

The HVAC auxiliaries formula, equation (B.5), has been adopted from previous study (Giechaskiel et al., 2021). T_{cabin} corresponds to a target cabin temperature in Celsius. The equation for cabin cooling applies to all powertrains, while the heating is operating for BEVs and OVC-HEVs while running in electric mode. The HVAC auxiliaries calculated via the equation are capped to a maximum value of 2.2 kW. The equations have been validated in previous studies (Komnos et al., 2022; Giechaskiel et al., 2021).

$$HVAC \text{ Auxiliaries [kW]} = \begin{cases} \frac{0.6 \cdot (T_{ambient} - T_{cabin})}{(25 - T_{cabin}) \cdot 1000}, & \text{for cabin cooling} \\ \frac{0.6 \cdot (T_{ambient} - T_{cabin})}{(13.5 - T_{cabin}) \cdot 1000}, & \text{for cabin heating} \end{cases} \quad (B.5)$$

OVC-HEV real-world CO₂ emissions

The real-world operation of OVC-HEVs requires extra considerations. A method analogous to the methodology to calculate the WLTP certified CO₂ values was used to weigh the CS and the entire electric operation. Each real-world trip is simulated both in full electric mode and CS mode. The CO_{2,CD,real-world} is calculated from the simulation as full electric; if it is possible to run the whole real-world trip in electric mode, then CO_{2,CD,real-world} is set to 0 g/km. If it is impossible, an average CO₂ emission as the CO_{2,CS,real-world} is considered for the remaining trip distance. Equation (B.6) presents the method as a formula. The combined CO₂ emissions are then calculated as in the WLTP, using a real-world UF (UF_{RW}) calibrated by real-world data (equation (B.7)).

$$CO_{2,CD,real-world} = CO_{2,CS,real-world} \cdot \frac{100 - \text{Trip electric percentage}}{\text{Trip distance}} \quad (B.6)$$

$$CO_{2,combined-weighted,real-world} = CO_{2,CS} \cdot (1 - UF_{RW}) + CO_{2,CD} \cdot UF_{RW} \quad (B.7)$$

Appendix C. Real-world trips insights

The 30 trips, 3 trips for each of the 10 distance groups, used for performing detailed vehicle simulations have been selected as representative ones only using the trip distance versus average trip speed domain, and performing K-means clustering algorithm (Komnos et al., 2022). Fig. 13 extends the analysis of the representativeness of the 30 trips by projecting additional characteristics of these 30 trips to the full real-world dataset. The trips used adequately cover the range of trip characteristics such as the average positive speed x accelerations, average positive and average negative

accelerations, stop percentage.

Regarding the full database's 483 trips' representativeness of real-world driving, a previous study (Suarez et al., 2023b) compared the trip distance versus average trip speed domain to another real-world campaign that took place during 2021–2022 and comprised 675 trips. The analysis showed a close agreement in the two databases' characteristics.

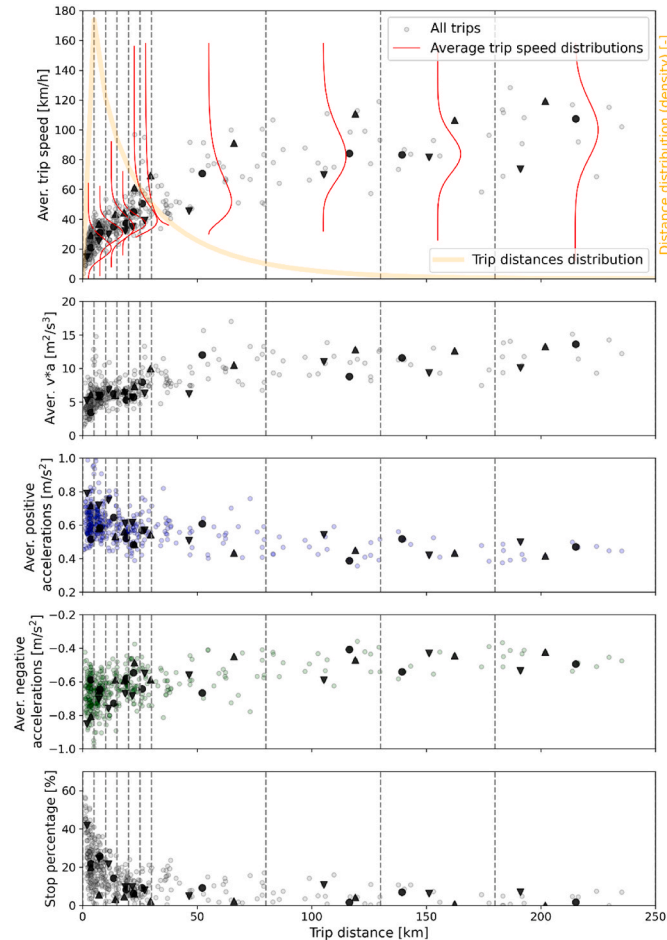


Fig. 13. Characteristics of the trips performed in the real-world campaign. Black markers in different shapes denote the trips selected for the simulation purposes. The vertical dashed lines denote the separation of the 10 distance groups.

Appendix D. Real-world monitoring

The European Commission (EC) collects data on the real-world CO₂ emissions, fuel or energy consumption of passenger cars and light commercial vehicles using on-board fuel and/or energy consumption monitoring (OBFCM) devices (European Union, 2019). Such data may be collected either through direct data transfers from vehicles to the manufacturers, or through their authorised dealers or authorised repairers. Since 2021, Pure ICE, NOVC-HEVs, and OVC-HEVs (PHEVs), using gasoline or diesel cars, are mandated to carry devices that capture the accumulated fuel/energy and distance (Regulation (EU) 2021/392). The accuracy of these devices is regulated and should not exceed $\pm 5\%$ of what is consumed. The EC recently published a detailed report on the topic (European Commission, 2024). The treatment and analysis of the EU_{RW,2021} dataset in the present study differs from the methodology followed in the EC report (European Commission, 2024). First, the records kept have minimum mileage of 1000 km, and secondly, the date of registration should be at least 100 days before December 2021. The two limits were imposed to limit the effect of higher engine frictions in a new car and get better representation of annual real-world driving.

To compare the gap calculated for the six countries selected to the simulations performed for same countries, the average distance travelled by each body style per Member State was used instead of the overall values presented in Table 2. The average distances per vehicle body are combined with the distance shares in urban, rural, and highway driving from the same table. The vehicle models were also simulated using the country's annual average – and extreme – ambient temperatures (more details in Appendix E).

Appendix E. Annual temperatures

Fig. 14 shows the distribution of the average annual temperatures for 1599 cities across the EU and 1196 areas in Australia (colour-filled distributions in figure) as collected from online sources²⁸. The cold period corresponds to months in Autumn and Winter, and hot period to months in spring and summer.

²⁸ Travel Weather Averages (<http://www.weatherbase.com/>).

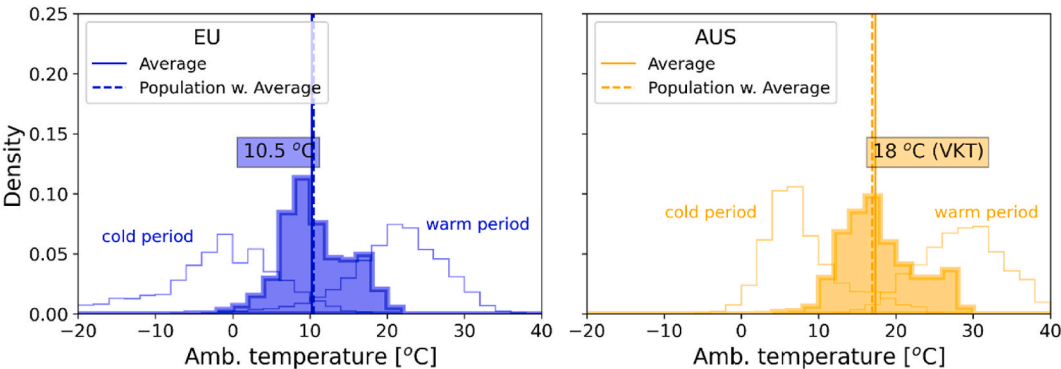


Fig. 14. Distributions of ambient temperature across EU (left) and Australia (AUS) (right).

Table E–1 provides the average ambient temperatures and the extremes selected for simulating the real-world CO₂ emissions for the 6 countries selected from the EU_{RW,2021} dataset (section 2.4.3).

Table E–1
Ambient temperatures used for the simulations performed for 6 EU Member States.

	Poland	Germany	France	Italy	Spain	Portugal
Average [°C]	7	8	11	14	16	16
Minimum [°C]	−8	−3	2	3	6	8
Maximum [°C]	24	21	24	27	29	26

Data availability

Data will be made available on request. A summary of the simulation data and results can be found in the JRC’s open data catalogue: <http://data.europa.eu/89h/5e327dfe-00a8-499e-be1e-bbc101e6e337>.

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